Searching Trainable Subspaces:
Masks and Codebooks for Finite-Budget Neural Optimization

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Abstract

Can structures recovered from solved tasks improve optimization on new tasks when the numerical source weights transfer poorly? We study a procedure that proposes masks or codebook assignments, trains fresh continuous parameters, and allocates further restarts to a selected structure. The analysis separates proposal coverage, conditional trainability, and candidate selection. For Gaussian sine prediction, we prove a finite-horizon separation between dense population gradient descent with fixed coordinate variance and search over covered, normalized low-dimensional subspaces. A stochastic extension requires additional amplitude-projection and minibatch assumptions. Controlled experiments compare frozen VAE generators and simpler priors at matched online training-example budgets. Source-derived sharing remains useful under a frequency shift that defeats numerical weight transfer. Without supplied coordinate groups, inferred components achieve 52.8% success at 32 weights versus 1.0% for tuned dense Adam; the same structures also support a stronger specialized non-gradient sine solver. For targets outside a proposed subspace, we derive its exact irreducible error and evaluate subsequent unrestricted optimization. At residual norm one, releasing weights yields 47.9% success while hard tying fails. Source corruption reveals differences between proposal mechanisms, whereas a realizable two-layer tanh control favors dense Adam over structural search. Together, these results characterize conditions and failure modes of transferring trainable parameterizations. They establish a conditional optimization benefit on synthetic problems; broad architectural generality, general VAE-specific superiority, and end-to-end efficiency remain open.

Keywords: subspace optimization; structural priors; weight sharing; codebooks; variational autoencoders; finite-budget optimization.

1 Introduction

An optimized network contains both numerical parameter values and structural information: which connections matter, which weights share a value, and which parts of a parameter vector vary together. These forms of information can transfer differently. A new task may preserve useful parameter-sharing relations while requiring numerical values far from those of the source solution. This motivates our question: when can structures extracted from solved tasks increase the probability of finding a low-risk predictor on a new task within a fixed online training budget?

We investigate *structural proposal-and-refit search*. A frozen generator proposes a discrete structure. Continuous parameters are optimized within that structure, candidates are compared on a selection sample, and the selected structure receives additional fresh initializations. A mask specifies which coordinates may vary. A codebook assignment specifies which coordinates must share a value. Both select a trainable subspace, although their representable function classes differ at a fixed parameter budget.

Our central finding is conditional. Transferred structure is useful when source solutions expose dependencies that persist on the target task, the proposal distribution covers appropriate combinations, and training within those combinations reaches accurate predictors often enough

to survive candidate selection. Approximate structure can also help: an initially restricted search may locate a useful region before all weights are released. A variational autoencoder (VAE), an empirical product over recovered parts, and enumeration provide different implementations of the proposal mechanism.

Weight sharing is already known to alter optimization difficulty (Shalev-Shwartz et al., 2017). The contribution here is a controlled account of *solution-derived structural transfer*: how useful parameterizations are recovered and recombined, where their finite search budget is lost, and what happens when the transferred equalities are only approximate. We develop three connected results:

1. **A mechanism with explicit conditions.** We express masks and normalized codebooks as trainable subspaces and separate proposal coverage, conditional optimization, and selection. A realizable Gaussian sine example admits finite-horizon probability separation from a specified dense population gradient method. We state the additional assumptions required for a stochastic extension and for source-profile identification (Sections 3 and 4).
2. **Structural transfer beyond numerical initialization.** On controlled mask and codebook tasks, recombining structures from learned source solutions improves success over tuned dense baselines. A frequency shift separates sharing relations from useful numerical warm starts. Hidden-group experiments remove supplied boundaries and identify source variation, coordinate correspondence, and search coverage as distinct requirements. Simpler priors and a specialized non-gradient solver locate which benefits belong to the representation rather than a particular VAE or conditional optimizer (Sections 6 to 8).
3. **Robustness and limits of approximate structure.** We derive the exact error floor imposed by a proposed subspace on Gaussian sine prediction. Experiments vary source quality and deviations from exact tying, and show when subsequent unrestricted training recovers useful solutions. A realizable two-layer tanh family provides a negative control in which dense Adam is already reliable (Sections 4.3 and 9).

The theorems establish sufficient conditions for specified algorithms and initializations. The experiments examine their proposed mechanism with separately tuned baselines; the actual minibatch runs do not satisfy every assumption of the stochastic theorem. The resulting claim concerns finite-budget optimization on the stated task families. It does not imply that dense optimization is impossible, that VAE proposals are universally preferable, or that source acquisition has been amortized. The experimental design in Section 5 distinguishes the information, initialization, and evaluation rules of each study before presenting its results.

2 Related Work

Subspaces and architecture search. Li et al. (2018) train networks in randomly oriented low-dimensional subspaces to measure the intrinsic dimension of objective landscapes. Paul et al. (2022) analyze information encoded in winning-ticket masks in terms of useful axial subspaces. Our question concerns transferring a distribution over structures and allocating a finite search budget on new tasks. Discrete structure selection followed by continuous training is also closely related to neural architecture search. DARTS uses a differentiable relaxation of architecture choices (Liu et al., 2018); our experiments use sampling and refits. We do not claim the general two-level optimization pattern as a new idea.

Weight sharing and codebooks. Most directly, Shalev-Shwartz et al. (2017) prove optimization separations between architectures with and without weight sharing, including an exponential time gap in a constructed setting. Both architectures can represent the target. Thus, the general claim that structural constraints can make a realizable problem easier for gradient descent is established prior work. Our question concerns the recovery and reuse of sharing relations

from learned source solutions. The experiments distinguish source identification, recombination, continuous fitting, and candidate selection, and evaluate approximate relations followed by unrestricted optimization. Our explicit separation calculation supplies a sufficient mechanism for this investigation; we do not claim priority for the general benefit of weight sharing. Soft weight sharing (Nowlan and Hinton, 1992) and description-length approaches (Hinton and van Camp, 1993) provide historical precedents for grouping and encoding network weights. Deep Compression explicitly stores weights using a codebook and indices, and fine-tunes shared centroids (Han et al., 2015). Here, assignments are transferred or recombined to define an optimization space for a new task; compression of an already trained target network is not the objective.

Generative optimization. VAEs (Kingma and Welling, 2013) and structured latent-space models such as Grammar VAE (Kusner et al., 2017) motivate learned proposal distributions. Generative models of network checkpoints (Peebles et al., 2022) and latent diffusion over parameters (Wang et al., 2024) address learning and transferring numerical parameters. Our principal controls distinguish this from transferring assignments and reinitializing their values. Finally, distribution-specific hardness of neural learning has substantial precedent (Shamir, 2018), including periodic ridge functions under Gaussian inputs and nearly flat objectives. Our finite-horizon sine calculation should be read as an explicit mechanism, not a priority claim for the broader phenomenon of weak gradient information.

3 Structural Proposal-and-Refit Search

For a task distribution $\mathcal{D}$, let

$$\mathcal{L}(w) = \mathbb{E}_{(x,y)\sim\mathcal{D}}\ell(f_w(x), y), \qquad w = \Phi_s(\theta), \qquad s \sim Q. \tag{3.1}$$

The discrete structure s selects a parameterization, while θ contains its trainable continuous parameters. A frozen VAE induces Q by sampling $z \sim \mathcal{N}(0, I)$ and discretizing its decoder output. No target-time gradient update is applied to the generator or to z in the principal series.

3.1 Learning what to transfer

The source stage first trains networks on source tasks and extracts discrete descriptions of their solutions. Successful masked fits supply masks; clustering learned weights supplies codebook assignments. A proposal distribution Q is fitted to these descriptions, or specified directly for uniform and enumeration controls. The numerical parameters of a new fit are then initialized afresh. This separates information about a useful parameterization from a numerical warm start, which we evaluate as a distinct control. Supplied role or block boundaries, inferred dependencies, and oracle target assignments provide different amounts of structural information; the experimental design identifies which methods receive each.

3.2 Two-stage allocation

The algorithm uses K exploratory fits and R refits:

1. Sample K structures from Q and train each for T minibatch steps.
2. Evaluate the trained candidates on a selection sample and retain the structure of the best candidate.
3. Train R fresh initializations within this selected structure.
4. Return the best candidate from both stages, using selection loss.

The online training budget, measured in example presentations, is

$$B_{\text{train}} = (K + R)Tb. \tag{3.2}$$

The principal setting is $K = R = 32$, $T = 1024$, and $b = 64$, giving 4,194,304 presentations per search. Dense restart baselines receive 64 independently drawn initializations. The original mask and codebook studies cycle four random initial vectors over the 32 structural proposals; the hidden-grouping, robustness, and two-layer studies instead use independent exploratory initializations. Fresh-value methods draw new refit initializations; numerical-transfer and fixed-value controls retain their stated alternatives. Table 1 records these differences. Training streams are shared within a context, so candidate outcomes need not be independent.

The *release* variant uses a fixed initial segment of each fit within the proposed subspace, then continues from the same weights with the tying constraints removed. Both exploration and refits follow this schedule. The total number of updates is unchanged; in the release experiments it is split into 768 restricted and 256 unrestricted updates. This uses a structure to guide the early optimization trajectory while restoring the full dense function class at the end.

For a structure s, let $b_T(s)$ be the probability that a complete fit reaches the desired risk, including initialization and training noise. The one-attempt probability is $p_Q = \mathbb{E}_{s\sim Q} b_T(s)$. This expression separates proposal coverage from trainability: a generator can frequently propose an appropriate subspace while its continuous fits still fail. Under independent complete attempts, the probability of obtaining an accurate candidate is $1 - (1 - p_Q)^N$; Appendix D states the relevant qualifications.

For two-stage allocation, let ρ_K denote the probability that exploration *selects* a good structure, and let β_T lower-bound the conditional success probability of a fresh refit. Under the conditional-independence assumptions stated in Proposition D.2,

$$\mathbb{P}(\text{an accurate candidate exists}) \geq \rho_K\big[1 - (1 - \beta_T)^R\big]. \tag{3.3}$$

If the probability that final selection discards all accurate candidates when one exists is at most δ_{sel}, the returned-success probability is at least

$$\Big[\rho_K\big(1 - (1 - \beta_T)^R\big) - \delta_{\text{sel}}\Big]_+ . \tag{3.4}$$

Seeing a good structure is therefore insufficient: an unsuccessful exploratory fit can prevent its selection, and final empirical selection can introduce another failure. These are conditional probability identities and bounds, not guarantees that a trained VAE attains the required coverage or that the experimental shared-data procedure satisfies independence. Our experiments report the risk of the *returned* predictor, using analytic population risk for sine tasks and separate evaluation samples for tanh tasks.

3.3 Masks and codebook assignments

A mask restricts a subset of weights to zero. For a scalar codebook containing k values, let $I \in \{0, 1\}^{k\times P}$ have one nonzero entry per column. With $U = I^\top$ and $c \in \mathbb{R}^k$, the parameterization is

$$w = Uc, \qquad w^\top = c^\top I, \qquad D = U^\top U = \operatorname{diag}(n_1, \ldots, n_k). \tag{3.5}$$

Empty codebook groups are removed. Thus a matrix product uses a one-hot assignment matrix, whereas an integer index array specifies the equivalent lookup. Codebooks constrain equalities between coordinates and can represent dense vectors using few trainable values.

Proposition 3.1 (Geometry of codebook updates)**.** *For $F(c) = \mathcal{L}(Uc)$, ordinary gradient descent induces*

$$\nabla_c F = U^\top \nabla_w \mathcal{L}, \qquad \nabla_c^2 F = U^\top (\nabla_w^2 \mathcal{L}) U, \qquad w^+ = w - \eta U U^\top \nabla_w \mathcal{L}. \tag{3.6}$$

Using the group-mean update instead gives

$$c^+ = c - \eta D^{-1} U^\top \nabla_w \mathcal{L}, \qquad w^+ = w - \eta P_U \nabla_w \mathcal{L}, \quad P_U = U D^{-1} U^\top. \tag{3.7}$$

Here P_U is the orthogonal projector onto range(U).

The proof is the chain rule and the projector identity; see Appendix B. With $V = UD^{-1/2}$, one has $V^\top V = I_k$ and $w = Va$. Isotropic initialization $a \sim \mathcal{N}(0, \sigma^2 I_k)$ corresponds to $c_j \sim \mathcal{N}(0, \sigma^2/n_j)$. Without this normalization, $c_j \sim \mathcal{N}(0, \sigma^2)$ yields $\mathbb{E}\,\|w\|^2 = \sigma^2 P$, even with a small codebook. The experiments vary initialization normalization and gradient summation separately.

4 Conditions for Effective Structural Search

We now give a sufficient mechanism for a structural advantage. The Gaussian example makes the risk and its gradients explicit, so representability can be separated from the probability of reaching an accurate solution. The result applies to a specified dense initialization and optimizer; subsequent experiments also evaluate tuned dense initializations beyond its assumptions.

4.1 Finite-horizon separation

Consider a realizable, one-output prediction problem

$$x \sim \mathcal{N}(0, I_P), \qquad y = \sin(\omega v^\top x), \qquad f_{\gamma,w}(x) = \gamma \sin(w^\top x), \qquad \|v\| = 1, \quad \omega > 0. \tag{4.1}$$

The loss is mean squared prediction error. The global minimum is zero at $(\gamma, w) = (1, \omega v)$. Rotate coordinates so that $v = e_1$, write $w = (t, u)$, and define

$$\mathcal{L}(\gamma, w) = A(w)\gamma^2 - 2B(w)\gamma + C_\omega, \tag{4.2}$$

$$A(w) = \tfrac{1}{2}(1 - e^{-2\|w\|^2}), \quad C_\omega = \tfrac{1}{2}(1 - e^{-2\omega^2}),$$

$$B(w) = \tfrac{1}{2} e^{-\|u\|^2/2} \left(e^{-(t-\omega)^2/2} - e^{-(t+\omega)^2/2} \right). \tag{4.3}$$

The target correlation is exponentially attenuated by the energy in orthogonal directions. This is a property of the actual prediction risk, not an auxiliary optimization objective.

Theorem 4.1 (Finite-horizon dense trapping)**.** *Let $|\gamma_0| \le 2$, $s = \|u_0\| - 1 \ge 1$, and $M = \|w_0\| + 1$. Define*

$$G(s, M) = 8Me^{-2s^2} + 4(1 + M)e^{-s^2/2}. \tag{4.4}$$

Joint population gradient descent with step $0 < \eta \le 1$ satisfies, whenever $\eta T G(s, M) \le 1$,

$$\|w_t - w_0\| \le 1, \quad |\gamma_t| \le 2, \quad \mathcal{L}(\gamma_t, w_t) \ge C_\omega - 3e^{-s^2}, \qquad 0 \le t \le T. \tag{4.5}$$

Inside the stated ball the weight gradient is small, while the amplitude update preserves $[-2, 2]$ as a convex combination. Induction closes the bound. This is finite-time persistence, not a claim of convergence to a bad local minimum. Appendix A gives the full proof and Gaussian tail constants.

For initialization $w_0 \sim \mathcal{N}(0, \sigma^2 I_P)$ with fixed $\sigma > 0$, define

$$s_P = \sigma\sqrt{(P-1)/2} - 1, \quad M_P = 2\sigma\sqrt{P} + 1, \quad \delta_P = e^{-(P-1)/16} + e^{-P/2}. \tag{4.6}$$

If $s_P \ge 1$, $\eta T G(s_P, M_P) \le 1$, and $3e^{-s_P^2} < C_\omega/2$, a dense run reaches risk at most $C_\omega/2$ with probability no greater than δ_P. The assumptions eventually hold for a fixed finite horizon as P grows. They do not give the same asymptotic result for fan-in variance proportional to $1/P$.

Theorem 4.2 (Covered trainable subspaces)**.** *Fix $\omega, \sigma > 0$ and $0 < \varepsilon < C_\omega/2$. Suppose a generator produces an orthonormal $V \in \mathbb{R}^{P\times k}$ with probability at least q such that $v \in \mathrm{range}(V)$ and $1 \le k \le k_0$, where k_0 is fixed. Initialize $a_0 \sim \mathcal{N}(0, \sigma^2 I_k)$ and $\gamma_0 = 1$, and optimize $\mathcal{L}(\gamma, Va)$. There are positive constants r, μ, Λ, independent of P and the subspace orientation, for which*

$$0 < \eta \le \min\{1, \Lambda^{-1}\}, \qquad \tfrac{1}{2}\Lambda r^2(1-\eta\mu)^{2T} \le \varepsilon \tag{4.7}$$

implies a one-attempt success probability of at least $q\beta_*$, where

$$\beta_* = \min_{1\le k\le k_0} \mathbb{P}_{Z\sim\mathcal{N}(0,\sigma^2 I_k)}(\|Z - \omega e_1\| \le r) > 0. \tag{4.8}$$

For N independent structural attempts, the probability that an accurate candidate exists is at least $1-(1-q\beta_*)^N$. Under the dense conditions above, N dense attempts have probability at most $N\delta_P$.

The reduction follows from $V^\top x \sim \mathcal{N}(0, I_k)$. At the exact solution the Hessian is positive definite, so a local convergence region has positive Gaussian initialization mass. Masks and normalized codebooks both satisfy this representation. Full details appear in Appendix B.

What must be paid for. The theorem requires coverage q; it does not learn this coverage for free. Even with a two-value codebook, a specific combination of J independent binary motifs has probability 2^{-J} under a uniform block prior. A small continuous dimension can therefore coexist with a difficult discrete search. The theorem matches coordinate variance in the *intrinsic* spaces, not the total ambient initialization norm. Our tuned dense experiments are not numerical certificates of this asymptotic theorem.

SGD and VAE-specific conditions. Appendix C controls dense escape by a stopped martingale and local structured convergence by an inexact-gradient coupling. It yields a strict probability separation for sufficiently large finite minibatches when the output amplitude is projected onto $[-2, 2]$. Actual experiments use batch 64 without projection; the sufficient batch condition is not established for them. Separately, a deterministic VAE decoder with reconstruction error e and aggregated-posterior divergence at most κ obeys

$$Q(\mathcal{G}) \ge P_{\text{source}}(\mathcal{G}) - e - \sqrt{\kappa/2}. \tag{4.9}$$

This bound requires additional assumptions for unseen whole structures and is not automatically certified by the training objective; see Appendix D.

4.2 When source solutions provide coverage

The separation theorem assumes a useful proposal distribution. A separate, elementary identification model explains how source variation can provide one. Suppose coordinates of each binary assignment are either constant or equal, up to fixed bit flips, to one of J latent factors. If every factor is fair, distinct factors are pairwise independent, and m source rows are independent and accurately recovered, exact matching of equal or complementary coordinate profiles recovers the varying groups with failure probability at most

$$\delta_m \le \min\{1, J(J+1)2^{-m}\}. \tag{4.10}$$

The proof and the thresholded and finite-bank variants appear in Appendix E. On successful recovery, independent uniform choice of each group's observed pattern assigns mass 2^{-J} to every combination, including unobserved complete assignments. This supplies a concrete route from

source identification to target coverage, while retaining an exponential combinatorial cost. A source family that never varies two factors separately can make them observationally indistinguishable.

This model assumes accurate assignment extraction. It does not analyze errors in trained source weights or establish a VAE learning guarantee. Section 8 examines source variation and correspondence, and Section 9 relaxes source accuracy empirically.

4.3 Approximate structure and unrestricted refinement

Exact target inclusion in a proposed subspace is a strong assumption. Let P_S be its orthogonal projector and decompose $w_\star = u + r$, where $u = P_S w_\star$ and $r = (I - P_S)w_\star$. For Gaussian inputs, Proposition F.1 gives the exact minimum risk among all predictors measurable with respect to $P_S x$:

$$R_S^* = \frac{1}{2}\left[1 - e^{-2(a+b)} - e^{-b}(1 - e^{-2a})\right], \qquad a = \|u\|^2, \quad b = \|r\|^2. \tag{4.11}$$

The conditional expectation $e^{-b/2}\sin(u^\top x)$ attains this minimum and belongs to the tied sine class. Thus optimizing longer within the same subspace cannot overcome the error floor. When a is large, the floor is approximately $(1 - e^{-b})/2$.

Approximate structure can nevertheless remain useful for initialization of unrestricted refinement. For any fixed predictor f, the risk under a target perturbation satisfies

$$\sqrt{R_{v+r}(f)} \le \sqrt{R_v(f)} + \|r\|, \qquad R_v(f) = \mathbb{E}[(f(x) - \sin(v^\top x))^2]. \tag{4.12}$$

Proposition F.2 proves this by the L^2 triangle inequality. It explains persistence of low-risk predictors under small perturbations, but does not guarantee that a perturbed training trajectory finds them. The release experiments in Section 9 measure when restricted optimization followed by unrestricted refinement overcomes this representation limit within a fixed budget.

5 Experimental Design

5.1 Questions and information available to each method

The experiments distinguish four potential sources of a structural advantage: restricting continuous optimization, transferring useful equalities rather than numerical values, recovering a proposal distribution from source solutions, and using an approximate structure before unrestricted refinement. The mask and codebook studies in Sections 6 and 7 establish the effect and isolate initialization and update geometry. Section 8 evaluates identification and search coverage without supplied coordinate groups. Section 9 tests source errors, approximate target tying, and a family on which dense optimization already succeeds.

Role-based mask priors and block-based codebook priors receive known boundaries. The hidden-group, robustness, and two-layer priors receive assignments extracted from learned source weights; their coordinate groups are inferred. Oracle controls alone receive the exact target assignment. Numerical-transfer controls reuse source parameter values, whereas structural methods initialize their continuous parameters afresh. All source-only generators are frozen before target optimization; CEM is an explicitly target-adaptive comparator. These information differences are part of the comparison, not interchangeable forms of access to the target solution.

5.2 Metrics, initialization, and comparison budget

Success always concerns the *returned* predictor. The sine studies use analytic population MSE below 0.1. The original tanh control uses a 0.1 threshold on 65,536 independent evaluation

Table 1: Protocol differences across search studies. Structural exploration has 32 fits, followed by 32 refits. Dense restart baselines have 64 independent initializations. Where used, evaluation samples are separate from the 4,096 examples used for candidate selection.

Study	Exploratory starts	Success criterion
Original masks and codebooks	Four initial vectors cycled over 32 proposals	Analytic population MSE < 0.1
Original tanh control	Four initial vectors cycled over 32 proposals	Evaluation MSE < 0.1; 65,536 examples
Hidden groups and robustness	32 independent initializations	Analytic population MSE < 0.1
Two-layer tanh	32 independent initializations	Evaluation MSE < 0.01; 16,384 examples

examples; the two-layer tanh study uses a 0.01 threshold on 16,384 examples in normalized target units. These thresholds define different study-specific criteria and their success rates are not pooled. Table 1 summarizes the differences. Neither evaluation risk nor diagnostic teacher parameters guide tuning, candidate selection, or learned-prior training. Oracle structures are the explicit exception for the named oracle controls and tied validation tuning in the two-layer study.

The principal stochastic search comparisons receive 64 fits of 1,024 steps at batch size 64, for 4,194,304 gradient-example presentations per search. The selection set has 4,096 examples independent of the training stream. The same set selects the structure for refitting and the final predictor; it is not an independent post-selection holdout. Exact population evaluation avoids using that set to estimate reported sine risk. Equal example presentations do not imply equal FLOPs or wall-clock time. Offline costs are reported separately.

The exact-population-gradient mechanism experiment evaluates individual initializations rather than this two-stage search. The specialized sine solvers in Section 8 count prediction evaluations, use additional model structure, and receive their own resource description. These controls are not included in an equal-compute claim.

For the original mask and codebook suites, dense optimizer settings are selected on separate validation functions by empirical success count, breaking ties by mean selected empirical loss. The mask suite tunes SGD, momentum, and Adam over learning rates and initialization scales. The codebook suite tunes SGD and Adam over rates and total initialization norms; the hidden-group and robustness studies retain these settings. The two-layer study separately tunes dense SGD, dense Adam, and tied SGD by mean empirical best-of-eight loss on new validation tasks. Appendix G lists the grids, chosen settings, and seeds. Shared minibatch streams induce dependence within a context. We therefore report counts, per-bank rates, and descriptive task-cluster intervals rather than treating every restart as an independent population replicate.

6 Structural Transfer with Masks

Masks provide the simplest instance of the mechanism: search chooses which coordinates are trainable. This study evaluates a structural advantage, the role of the generator, and the effect of adding known nuisance inputs.

Table 2: Primary mask comparison. Each method receives 384 searches at the same online training-example budget. Success means returned population MSE < 0.1. Role-based priors and enumeration receive the known role partition.

Method	Successes	Success rate	Mean MSE
Oracle mask	383/384	99.7%	0.0541
Enumerate 16 role-compatible masks	311/384	81.0%	0.1002
Uniform over known roles	269/384	70.1%	0.1282
Partwise VAE	259/384	67.4%	0.1347
Empirical product	251/384	65.4%	0.1398
Dense Adam	103/384	26.8%	0.2251
Uniform over all two-edge masks	49/384	12.8%	0.2722
Vanilla VAE	29/384	7.6%	0.2855
Dense momentum	16/384	4.2%	0.2976
Dense SGD	6/384	1.6%	0.3269
Whole-mask elite replay	0/384	0.0%	0.3045

6.1 Tasks and source banks

The mask family uses two labeled outputs:

$$x \sim \mathcal{N}(0, I_d), \quad y_j = \alpha_j \sin(\ell_j x_{r_j}) + \sin(h_j x_{r_j}), \quad f_j(x) = \gamma_j \sin(w_j^\top x). \tag{6.1}$$

Here $\ell_j \in [0.65, 0.75]$, $h_j \in [3.8, 4.2]$, and $\alpha_j \in [0.35, 0.45]$. At $d = 8$, the first relevant coordinate belongs to the first four inputs and the second to the last four. A mask keeps two of 16 connections; output amplitudes remain trainable. There are 120 two-edge masks and 16 that respect the known roles.

Source tasks cover eight pairs of parts, with four other pairs each assigned to validation and evaluation. All individual parts are observed in source tasks, but complete evaluation pairs are withheld. The original source bank contains 279 elite solutions from 262,144 random-mask fits. The primary comparison evaluates three VAE training seeds, eight task/function contexts, and 16 search trials per context, giving 384 searches per method. A second series collects three new source banks, trains one new partwise VAE per bank, and evaluates new functions after within-role coordinate permutations.

The two-frequency teacher is not exactly representable by the one-sine student. We do not assert that its observed low-risk solution is the global minimum. The realizable theoretical prototype and the codebook family below have explicitly known zero-risk solutions.

6.2 Matched-budget comparison

Table 2 shows a substantial improvement over the tested dense optimizers, but no special advantage of the VAE over simple structural proposals. Enumeration is strongest among non-oracle methods, consistent with the small space of 16 role-compatible masks. Whole-mask replay fails on withheld combinations.

The VAE–product difference is +2.1 percentage points, with a descriptive 95% task-cluster bootstrap interval $[-2.9, 7.3]$. The corresponding difference against dense Adam is +40.6 points, with interval $[27.1, 52.6]$. Resampling uses eight function/family clusters and is conditional on the original bank; it is not a generalization guarantee or a multiple-comparison correction. Additional paired differences appear in Appendix G.

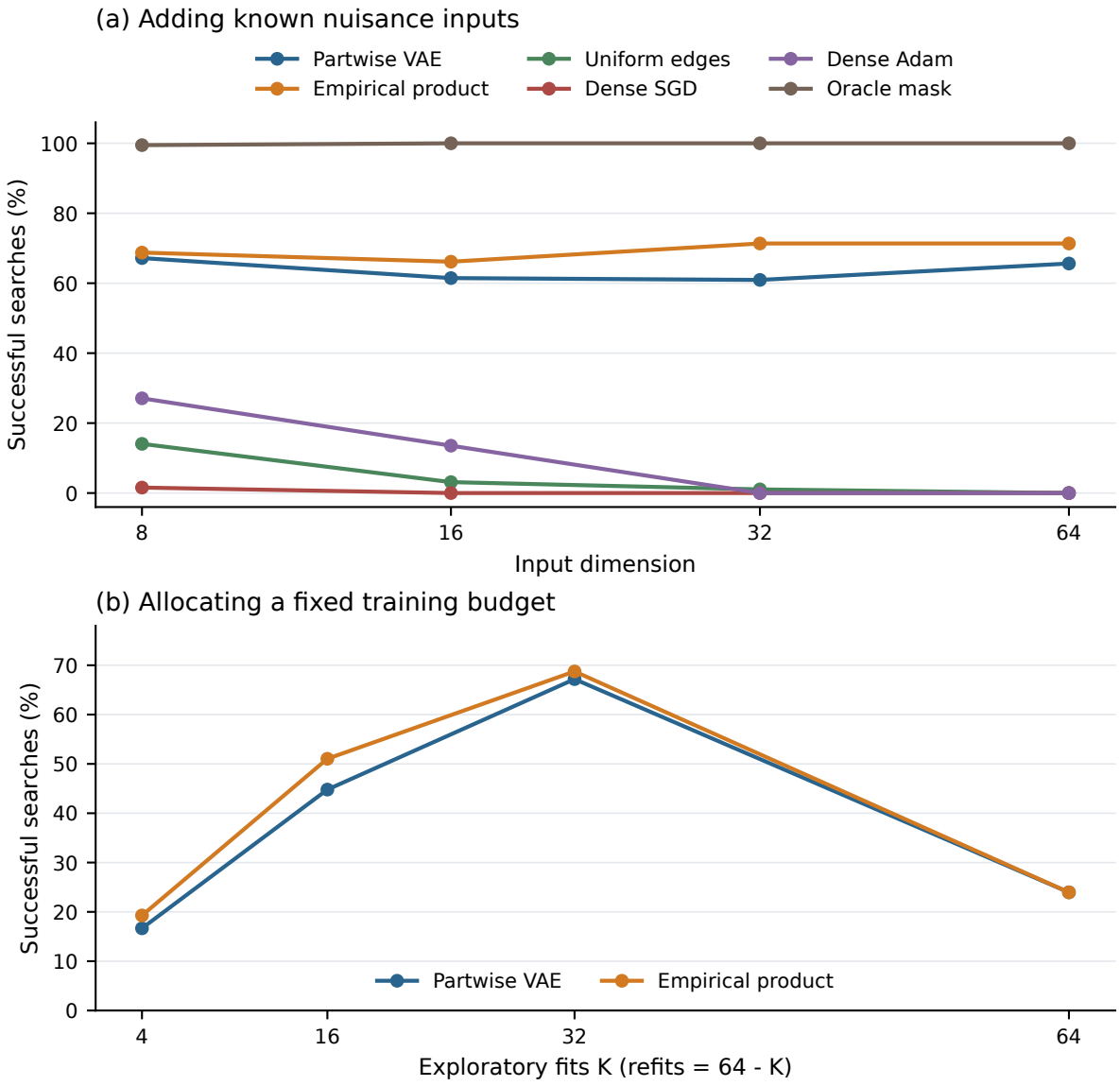


Figure 1: Mask search under added known nuisance coordinates (a) and alternative exploration/refit allocations (b). Each point aggregates 192 searches. Dense optimizers are tuned separately at each dimension.

6.3 Dimension, allocation, and independent banks

Adding known irrelevant inputs up to $d = 64$ leaves the partwise VAE at approximately 61–67% success and the product prior at 66–71%. Dense Adam declines from 27.1% at $d = 8$ to no observed successes among 192 searches at $d = 32$ and $d = 64$. The frozen structural prior is explicitly padded with zeros outside the original inputs: this is a nuisance-coordinate control, not a demonstration of discovering unknown support at larger d.

At a fixed budget of 64 fits, the allocation $32 + 32$ outperforms the tested $4 + 60$, $16 + 48$, and $64 + 0$ choices (Figure 1). Exploration always cycles four weight initializations, so the $64 + 0$ result does not characterize pure sampling with 64 independently initialized fits. The $32 + 32$ allocation and $d = 8$ scaling observations share the same design and seeds and are not independent replications.

On three independently collected banks, partwise VAE search achieves 247/384 successes (64.3%), empirical product 256/384 (66.7%), role-uniform sampling 265/384 (69.0%), enumera-

tion 317/384 (82.6%), and dense Adam 139/384 (36.2%). VAE rates range from 60.9% to 66.4% across banks. These results reproduce the structural effect, while retaining the finding that simple generators are competitive. One VAE is trained per bank, so bank variation and VAE training variation are not disentangled.

6.4 Mechanism and negative controls

In the realizable population experiment, a correct one-coordinate mask retains about 59% success per individual initialization as dimension grows to 128. Dense fixed-scale GD drops from 1.33% at $d = 8$ to 0/8192 at $d \geq 16$. The actual sufficient trapping condition is also evaluated for each initial point (Table 12). This experiment uses exact population gradients, not minibatch SGD.

A separate representation control trains categorical VAEs on eight explicitly provided diagonal structures. With eight dependent parts, the average prior mass per target structure is about 0.12, versus 6.68×10^{-8} for the empirical marginal product and 0.125 for replay. This demonstrates dependency modeling in the supplied structure distribution, not unseen-structure transfer or an additional end-to-end solver result.

Finally, on a realizable tanh family, dense SGD achieves 100% at $d = 8$ and 96.9% at $d = 32$, compared with 83.3% for the frozen sine-trained VAE in both cases. This changes both the activation and the teacher family; it is not an isolated activation ablation. It demonstrates that external structural search can be detrimental when direct training is already reliable.

7 Transferring Codebook Structure

Codebooks extend structural transfer to dense target vectors that sparse masks cannot represent. The key comparison holds the assignment mechanism fixed while changing whether numerical codebook values are transferred or initialized afresh. Additional controls isolate initialization normalization and the metric induced by sharing.

7.1 Learning assignments from easier source problems

We now use a realizable dense teacher with two repeated weight values:

$$y = \sin((w^*)^\top x), \qquad w_p^* = \frac{\omega}{\sqrt{P}}(2i_p^* - 1), \qquad i_p^* \in \{0, 1\}. \tag{7.1}$$

The index vector consists of J known blocks of width eight. Each block is one of the balanced motifs 11110000 and 11001100. Source tasks contain all even-parity combinations of motif indices and frequencies in $[0.8, 1.2]$. Evaluation uses unseen odd-parity combinations and frequencies in $[3.8, 4.2]$. We study $J = 2, 4, 6$, corresponding to $P = 16, 32, 48$ weights.

For each of three bank seeds per dimension, source solutions are trained from random dense initialization by ordinary SGD. Four restarts are used per task. Solutions with empirical selection MSE below 0.005 are retained, their amplitude sign is canonicalized, and two scalar centroids are fitted to their learned weights. The generator receives the recovered assignments, not teacher assignments. All nine banks recover source assignments exactly; this accuracy is measured only as a diagnostic and is not the acceptance rule.

A block VAE is trained on the recovered eight-bit blocks and then frozen. It samples blocks independently at target time. The empirical-block control samples recovered source blocks at each block position. Both receive known block boundaries. Independent-bit and whole-assignment replay controls remove within-block dependence or the ability to form new whole combinations, respectively. The main codebook method reinitializes two trainable centroids and an output amplitude on every fit; normalized intrinsic coordinates have standard deviation

Table 3: Codebook search success rates (%). The denominator per method is 192, 768, and 768 for 16, 32, and 48 weights, respectively. All methods use 64 fits per search. Oracle assignments expose target structure; other proposal methods use only recovered source assignments.

Method	16 weights	32 weights	48 weights
VAE assignments, new random values	87.0	49.1	12.8
Empirical blocks, new random values	87.0	46.6	17.4
Independent bits, new random values	3.1	0.1	0.0
Whole-assignment replay, new values	0.0	0.0	0.0
Empirical blocks, transferred values	0.0	0.0	0.0
Empirical blocks, fixed source values	0.0	0.0	0.0
Empirical blocks, unnormalized values	65.6	19.1	4.8
Empirical blocks, summed gradients	84.4	41.1	13.3
Oracle assignments, new random values	100.0	100.0	100.0
Oracle assignments, transferred values	0.0	0.0	0.0
Dense SGD	10.4	0.0	0.0
Dense Adam	33.3	1.2	0.0
Dense source-weight replay	0.0	0.0	0.0
Two-edge sparse mask	0.0	0.0	0.0

two, giving expected squared ambient norm eight when both groups are used. Default updates average gradients within each group.

7.2 Transfer of structure versus transfer of values

Table 3 reports 192 searches per method for $P = 16$ and 768 per method for each larger dimension. At $P = 32$, VAE assignments succeed in 377/768 searches (49.1%), empirical blocks in 358/768 (46.6%), and dense Adam in 9/768 (1.2%). Dense SGD has no observed success at $P = 32$ or 48.

Correct assignments with fresh random codebook values succeed in all 1,728 observed searches across dimensions. The same assignments initialized with source values yield no successes. Thus the tested frequency shift separates useful parameter sharing from useful numerical initialization. It does not imply that warm starts are generally harmful. Dense replay of complete source weight vectors also fails in this setting.

The representation controls require care. A two-edge mask cannot represent the dense teacher exactly; Proposition B.2 quantifies the resulting irreducible error. Likewise, fixing source codebook values restricts available frequencies. These are controls for representability and transfer, not equal-capacity optimization competitors.

7.3 Normalization and the remaining search bottleneck

At $P = 32$, removing centroid initialization normalization lowers empirical block success from 46.6% to 19.1%. Replacing mean gradients with summed gradients, while retaining normalized initialization, lowers it to 41.1%. Each control changes one factor, consistent with Proposition 3.1.

Coverage degrades with the number of combinations. At six blocks, VAE search encounters correct assignments in 36.6% of searches but selects them for refitting in only 12.8%. Empirical blocks encounter them in 41.5% and select them in 17.4% (Table 4). Both proposal coverage and exploratory assessment limit performance, despite perfect observed search success with oracle assignments. No consistent VAE advantage over empirical blocks is established.

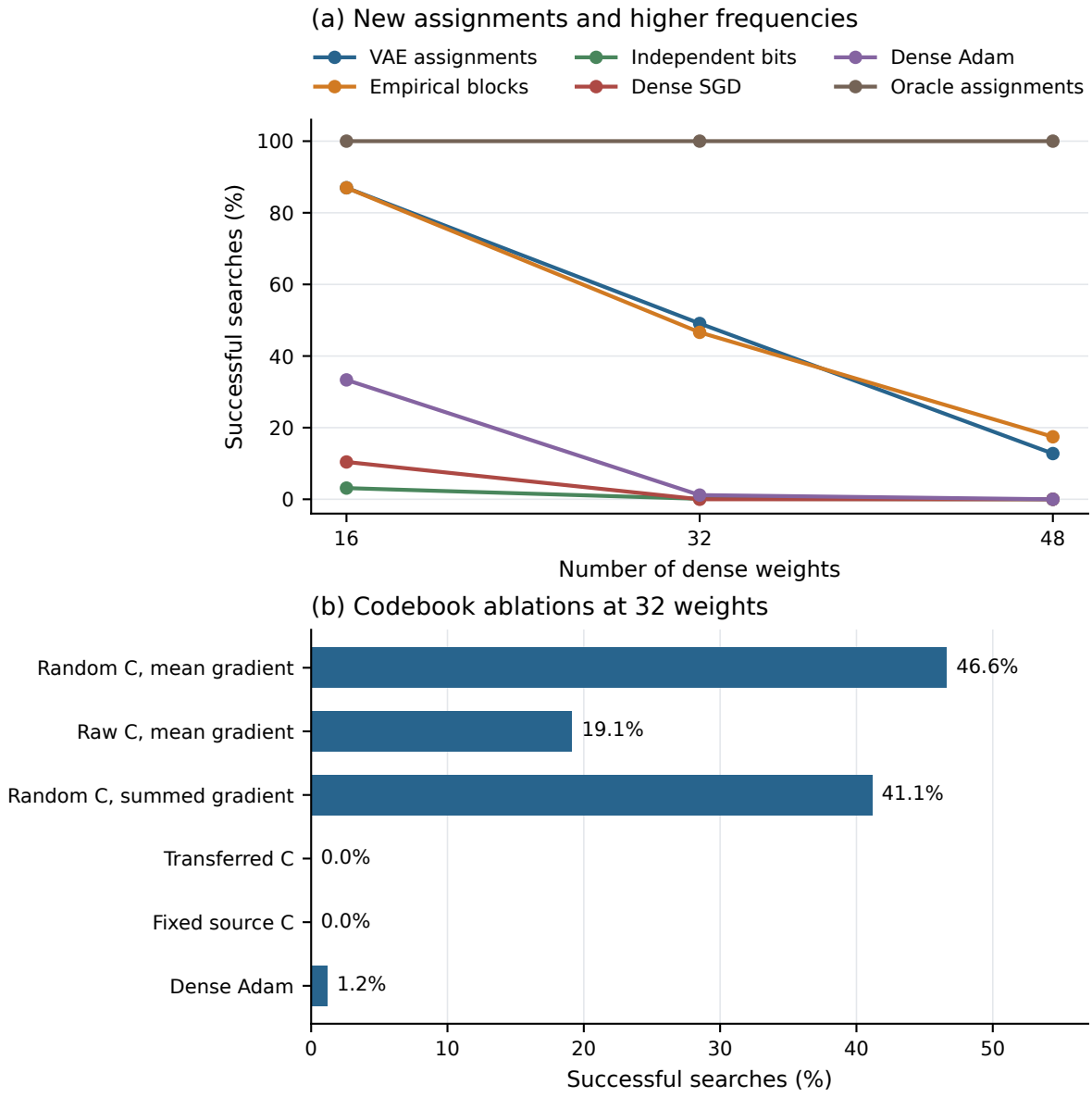


Figure 2: Codebook search across ambient dimensions (a) and separate initialization/gradient-metric ablations at 32 weights (b). A small continuous codebook does not eliminate the cost of finding its assignments.

8 Recovering Proposal Structure from Source Solutions

The codebook experiment separates structure from numerical initialization, but supplies the boundaries used for recombination. Here we ask whether learned source solutions reveal useful dependencies when those boundaries are hidden, and whether the recovered proposal support benefits different search algorithms. The frozen protocol reuses the three trained source banks per dimension, applies a different hidden coordinate permutation to each bank, and evaluates fresh high-frequency target tasks with held-out odd-parity combinations. The learner receives only assignments extracted from the optimized source weights. The target frequencies use new seeds, and the dense settings remain fixed from the earlier independent validation tasks.

Learning dependencies. A full-vector categorical VAE is trained without a supplied partition. A simpler comparator groups varying coordinates whose source profiles agree or disagree

Table 4: Correct-assignment diagnostics in the first 32 codebook fits. Global exchange of the two codebook labels is treated as equivalent. "Selected" refers to the assignment retained for the refit stage.

Blocks	VAE encountered	VAE selected	Blocks encountered	Blocks selected
2	100.0%	87.0%	100.0%	87.0%
4	87.5%	49.1%	87.4%	46.6%
6	36.6%	12.8%	41.5%	17.4%

Table 5: Hidden-grouping follow-up with full source banks and shared coordinate semantics. Stochastic methods have 144, 288, and 288 searches at 16, 32, and 48 weights, respectively. Specialized non-gradient controls have 12, 24, and 24 task-context evaluations. The latter use a different solver and resource unit; their rows are separated accordingly.

Method	16 weights	32 weights	48 weights
Full-vector VAE	0.0%	38.5%	15.3%
Inferred components	0.0%	52.8%	17.0%
Known source blocks (privileged)	93.8%	54.9%	18.4%
Wrong block boundaries	0.0%	0.7%	0.0%
Independent source marginals	7.6%	0.0%	0.0%
Whole-assignment replay	0.0%	0.0%	0.0%
Uniform assignments	0.0%	0.0%	0.0%
Enumerate inferred support	0.0%	55.6%	13.2%
CEM over inferred components	0.0%	44.8%	17.7%
Coordinatewise CEM	3.5%	0.0%	0.0%
Exact assignments (privileged)	100.0%	100.0%	100.0%
Dense SGD	8.3%	0.0%	0.0%
Dense Adam	38.2%	1.0%	0.0%
Spectral / inferred support	0.0%	100.0%	100.0%
Spectral / uniform assignments	0.0%	0.0%	0.0%
First moment + radial search	0.0%	0.0%	0.0%

on at least 95% of observations, then resamples observed patterns independently across the inferred components. Constant coordinates form a fixed component. This recovers statistical dependencies, which need not coincide with the original geometric blocks. Known permuted blocks are retained as a privileged comparator; consecutive coordinates after permutation provide a wrong-boundary control. Further controls use independent source marginals, replay, uniform assignments, and exact target assignments. Appendix E gives an elementary sufficient condition for source-profile identification and explains the two-block failure case.

Budget and adaptive search. Every stochastic search uses 32 exploratory fits and 32 fresh refits, each with 1,024 updates of batch size 64. All exploratory continuous initializations are independent. This differs from the repeated four-start exploration in the earlier codebook study, so the two studies are not a single-factor ablation. Enumeration uses the inferred product support in random order, cycling it when smaller than 32, with independent initializations on repeats. For larger support it draws 32 distinct structures. Categorical cross-entropy search (CEM; de Boer et al., 2005) instead spends four rounds of eight fits, updating proposal probabilities from two empirical-loss elites with fixed smoothing. We compare CEM over inferred components with coordinatewise CEM. All candidates share training streams and a separate 4,096-example selection sample within each context; analytic population risk never guides search.

Table 6: Source-size and correspondence controls. Each stochastic entry uses 288 searches; each specialized entry uses 24 task contexts. The final column uses the full source bank but a fresh target-coordinate permutation.

	Eight source solutions		Broken transfer
Method	32 weights	48 weights	48 weights
Full-vector VAE	27.4%	1.7%	0.0%
Inferred components	51.7%	12.2%	0.0%
Known source blocks (privileged)	48.3%	9.4%	0.0%
Enumerate inferred support	55.6%	9.4%	0.0%
CEM over inferred components	44.1%	9.4%	0.0%
Coordinatewise CEM	0.0%	0.0%	0.0%
Exact assignments (privileged)	100.0%	100.0%	100.0%
Dense Adam	1.0%	0.0%	0.0%
Spectral / inferred support	100.0%	75.0%	0.0%
First moment + radial search	0.0%	0.0%	0.0%

Specialized non-gradient comparators. The spectral comparator enumerates up to 64 inferred structures and, for each, fits a single sine along its normalized sign direction. It scans 64 radial frequencies from 0.125 to 8 and computes the least-squares output amplitude on 1,024 training examples. Frequency is chosen by training error and the returned structure by the independent selection sample. A second version uses uniform assignments. A source-independent first-moment method estimates a direction from the empirical mean of $x \sin(w_\star^\top x)$ using the same 65,536 unique training examples as the stochastic fits, then performs radial search. These solvers use no gradient to find weights.

The structural spectral methods spend 4,194,304 prediction-example evaluations, whereas the stochastic searches spend that number of gradient-example presentations. The spectral solver also uses the single-sine, antipodal-value model assumption. Consequently, these are informative specialized competitors, not equal-FLOP or equal-latency measurements. Their deterministic output is counted once per task context, rather than replicated twelve times alongside the stochastic searches.

Source size and correspondence. We compare nested subsets of eight accepted source solutions with the full banks, omitting redundant subset conditions at two blocks. A negative control uses six blocks and full banks but gives each target a fresh coordinate permutation, breaking source–target correspondence. There are 132 task contexts and 20,988 returned solutions across all methods. Twelve stochastic searches in a context share data and are not twelve independent tasks; there are only three source-bank replications. Raw counts and bank-level rates accompany the manuscript.

Interpretation. At 32 weights, inferred-component sampling succeeds in 152/288 searches (52.8%), compared with 158/288 (54.9%) for known blocks, 111/288 (38.5%) for the full-vector VAE, and 3/288 (1.0%) for dense Adam. At 48 weights, component sampling and the VAE reach 17.0% and 15.3%, respectively. The tested CEM configuration does not consistently improve on independent component proposals. These observations support source-based recovery of a useful parameterization, without establishing a VAE-specific advantage.

The strongest positive comparator is specialized spectral search over the inferred support: it solves all 24 evaluated contexts at each of 32 and 48 weights with full banks. Spectral search over uniform assignments and the first-moment direction method solve none. Thus, the recovered structure is useful, while gradient-based conditional refitting is not the strongest solver for this

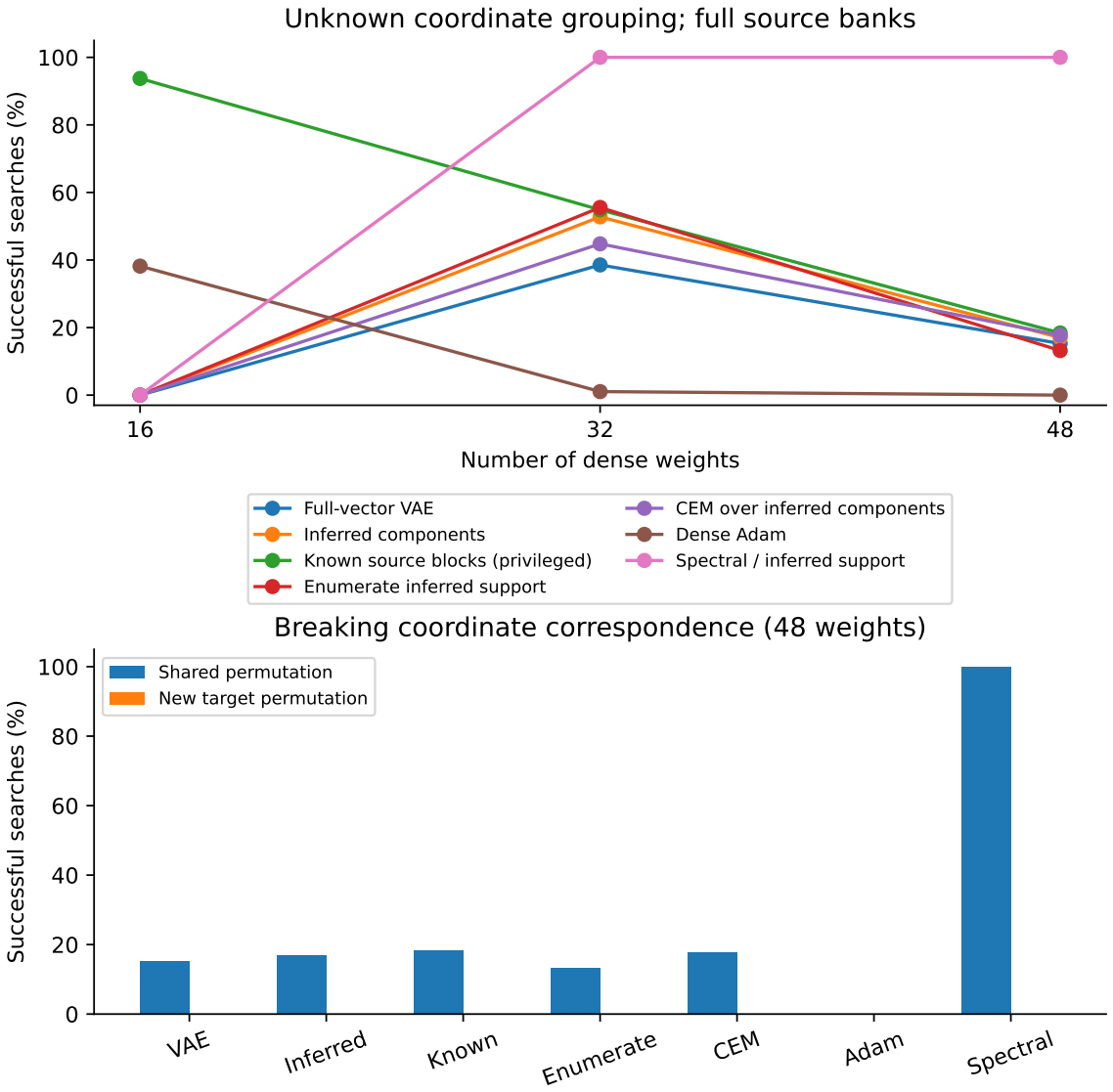


Figure 3: Hidden-grouping follow-up. Top: full-bank success under a shared hidden coordinate permutation. Bottom: a fresh target permutation breaks transfer correspondence. Specialized spectral results use a different computational resource unit and extra single-sine model knowledge.

controlled sine family. The computational units and extra model knowledge differ as described above.

The controls delimit the claim. At two blocks, source profiles merge the varying factors and inferred-component search fails, while supplied boundaries yield 93.8% success. With eight source solutions at six blocks, one of the three banks merges two factors, and spectral success falls to 18/24 contexts. When target coordinates are independently permuted, all transferred-search methods have zero observed successes, while exact assignments still yield 288/288. These are source-identification and correspondence requirements, not properties guaranteed by the generator architecture.

Full-bank profile recovery relies on accurate binary assignments and factor variation. The inferred support still has exponentially many combinations as the number of varying factors grows. Specialized enumeration at these dimensions therefore does not establish scalability. The VAE and CEM results concern the fixed configurations in the protocol, not all possible variants.

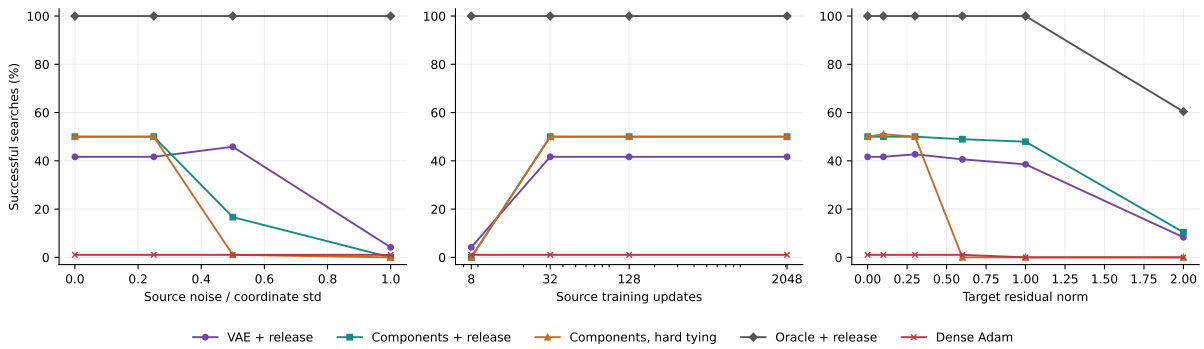


Figure 4: Robustness at 32 input weights. Each point aggregates 96 searches across three source banks. The panels independently vary source noise, source-training duration, and target residual norm. The 2,048-update point uses the original accepted source bank. Curves are descriptive; bank-specific rates, continuous errors, and task-context bootstrap intervals are retained in the machine-readable summary.

9 Robustness and Limits of Structural Transfer

Accurate recovery and exact target inclusion favor structural transfer. This section tests how the benefit changes when source solutions are imperfect, target weights violate the inferred equalities, or dense training is already effective. The source-quality and residual-size sweeps measure the first two conditions; a realizable two-layer tanh family tests the third. Each study follows a protocol frozen before evaluation.

9.1 Source quality and approximate weight sharing

At $D = 32$, we reuse three source banks and evaluate two new frequency seeds with two predetermined held-out motif combinations. Each condition contains eight searches in each of 12 task/source-bank contexts, or 96 returned solutions per method. We add Gaussian perturbations to learned source weights with standard deviation $\sigma \in \{0, 0.25, 0.5, 1\}$ times their coordinate standard deviation, then re-extract assignments and retrain the priors. A separate curve uses actual dense source solutions after 8, 32, or 128 updates. These retain every source task after empirical best-of-four selection, without filtering inaccurate solutions.

For approximate tying, write $w_\star = w_0 + \epsilon r$, where w_0 follows the original target assignment, and r is a unit vector orthogonal to both assignment indicators. Clean source banks are evaluated at $\epsilon \in \{0, 0.1, 0.3, 0.6, 1, 2\}$. The additional predeclared interaction uses $\sigma = 0.5$ and $\epsilon = 0.3$. The resulting 13 conditions yield 156 contexts and 9,984 returned solutions across eight methods.

Each search uses 32 exploration fits and 32 fresh refits, each with 1,024 updates of batch size 64. Release methods keep the structure for 768 updates and then optimize all weights for 256 updates, preserving the current weights and amplitude. Hard tying uses the entire budget within the subspace. Dense SGD and Adam are unrestricted throughout and reuse the original independent validation settings. The component methods with and without release share the same proposals and initial values. Selection uses 4,096 independent examples; success remains analytic population MSE below 0.1. Matching 4,194,304 gradient-example presentations per search does not match FLOPs or end-to-end costs.

With clean sources and exact tying, components followed by release succeed in 48/96 searches (50.0%), versus 40/96 (41.7%) for the VAE and 1/96 (1.0%) for dense Adam. Source noise $\sigma = 0.25$ leaves every extracted assignment unchanged. Source training for only 32 updates also recovers all assignments, even though one bank's mean source risk is approximately 0.0094; the search outcomes are identical because the generators consume assignments rather than numerical source values. At eight updates, source assignment bit errors reach 14.5–18.2%, component

search has no observed success, and the VAE succeeds in only 4/96 searches.

Moderate source noise exposes a difference between the priors. At $\sigma = 0.5$, only 1.6–3.5% of source assignment bits are incorrect, but the thresholded dependency extractor fragments the factors: its support grows from 16 combinations to 4,096, 8,192, or 131,072, depending on the bank. The released component method falls to 16/96 successes (16.7%), while the released VAE succeeds in 44/96 (45.8%). A post-run coverage diagnostic finds the exact base assignment among the first 32 proposals in 9/96 component searches and 55/96 VAE searches, respectively. Release can also rescue an inexact assignment, so coverage and returned-solution success are distinct. This is a specific robustness advantage over this dependency extractor, not a comparison with all noise-aware non-neural priors. At $\sigma = 1$, both methods deteriorate sharply: 0/96 for components and 4/96 for the VAE.

With clean sources, approximate tying is tolerated when weights are subsequently released. At residual norms $\epsilon = 0.6$ and 1, released components succeed in 47/96 (49.0%) and 46/96 (47.9%) searches, respectively, while hard tying succeeds in none. Dense Adam reaches 1/96 and 0/96 in these conditions. At $\epsilon = 2$, released components fall to 10/96 (10.4%), and even the oracle base assignment followed by release falls to 58/96 (60.4%). The oracle succeeds in all 96 searches at every smaller residual. The predeclared combined disturbance $(\sigma, \epsilon) = (0.5, 0.3)$ yields 41/96 VAE and 17/96 component successes. Thus release substantially enlarges the useful regime, but sufficiently large source or target disturbances still defeat the finite search budget.

The behavior under target perturbations is consistent with the error floor in Section 4.3. With the correct base assignment and $\|w_0\| \simeq 4$, the floor is approximately $(1 - e^{-\epsilon^2})/2$: it already exceeds the success threshold at $\epsilon = 0.6$. Releasing weights removes this representation restriction. The experimental gain measures the ability of the released trajectory to escape that restriction; the theoretical error floor itself does not guarantee convergence of unrestricted refinement.

9.2 When dense optimization already succeeds

We next consider

$$f_\theta(x) = \frac{1}{\sqrt{H}} \sum_{h=1}^{H} a_h \tanh(w_h^\top x + b_h) + b_0, \qquad H = 4, \quad D = 32. \tag{9.1}$$

The teacher and dense student have the same 137-parameter architecture. The target input weights share a binary coordinate partition, but each hidden unit has its own two-value normalized codebook. With the correct partition the 17-parameter tied model represents the target exactly. The output weights and biases are trainable in every method.

Each of three source banks contains 16 easy tasks: all eight even-parity motif combinations with two sets of numerical coefficients. Four dense Adam restarts per task run for 4,096 updates. We select by empirical error and retain all tasks. Two-means clustering of the columns of each learned input-weight matrix extracts the assignment. Column distances are invariant to permutations and sign changes of hidden units, so this extraction does not require matching hidden-neuron identities across source solutions. The generator receives extracted assignments and no teacher parameters. Every extracted source partition is correct in this study; mean source evaluation MSE is below 10^{-3} in each bank.

New targets use odd-parity combinations and fresh hidden/output coefficients and biases. The standard deviation of normalized input codebook values changes from 0.8 to 2.5. Targets are centered and scaled using an independent 8,192-example calibration sample. Dense SGD, dense Adam, and tied SGD are separately tuned over three initial norms and three learning rates on two additional validation tasks. Tied tuning uses the correct validation partition, a disclosed privilege shared by all structural methods. Evaluation outcomes are not used for tuning.

Method	Success (%)	Mean MSE	Bank range (%)
VAE	75.0	0.0899	71.9–78.1
Components	81.2	0.0569	78.1–84.4
Components + release	83.3	0.0076	75.0–90.6
Partition replay	0.0	0.3069	0.0–0.0
Uniform	0.0	0.4880	0.0–0.0
Oracle partition	100.0	0.0017	100.0–100.0
Dense SGD	75.0	0.0103	75.0–75.0
Dense Adam	100.0	0.0005	100.0–100.0
Numerical transfer + Adam	100.0	0.0005	100.0–100.0

Table 7: Two-layer tanh results: 96 searches per method, clustered in 12 task/source-bank contexts. The final column shows the minimum and maximum success percentages among the three source banks. Each bank has 32 searches. Mean MSE uses a held-out evaluation set.

The evaluation again has 12 contexts and eight searches per context, with 32 exploration fits and 32 refits of 1,024 updates each. Success is MSE below 0.01 on a separate 16,384-example evaluation sample; this is an empirical estimate rather than exact population risk. In addition to the structure controls, we include full source-parameter transfer followed by dense Adam, using fresh small perturbations of the source network for each restart.

Dense Adam and numerical transfer solve all 96 observed searches; dense SGD solves 72. The VAE solves 72, inferred components 78, and components followed by release 80. The oracle partition solves all 96, while source-partition replay and uniform assignments solve none. Accurate source identification therefore does not suffice to make structural search the best optimizer: the dense problem is already accessible to Adam, and proposing and selecting structures introduces failures. The additional layer supplies a useful negative control, not evidence of a universal structural advantage.

Source acquisition for each two-layer bank costs 16,777,216 gradient-example presentations, equivalent to four target-search budgets before prior training and validation tuning. Joint vectorized worker times cannot be interpreted as per-method latency. Across both follow-ups we retain all 10,848 returned models' errors, candidate-selection indices, source diagnostics, and runtime source snapshots. Only three source banks and four numerical target/motif contexts per bank are evaluated; resampled context intervals are descriptive and do not establish broad distributional generalization.

10 Discussion and Limitations

What transfers, and when. The results support transferring a trainable parameterization across tasks whose useful numerical weights can differ substantially. Masks restrict coordinates; codebooks preserve dense directions with few free values. The frequency-shift control isolates this distinction: the correct sharing assignment with fresh values succeeds, while the same assignment initialized with source values fails under the tested training budget. The hidden-group study shows that supplied block boundaries are not essential when source variation makes dependencies identifiable. Its failures at two factors and under a fresh target permutation identify the corresponding source-variation and coordinate-correspondence requirements.

Where the search budget is lost. Recovering a useful decomposition does not ensure that a finite search finds and selects the right combination. The coverage diagnostics separate seeing a correct assignment from selecting it for refitting; oracle controls isolate continuous fitting after

that uncertainty is removed. The specialized sine solver further shows that a learned proposal support can remain useful when the conditional optimizer changes. On that family it is stronger than our gradient-based refits, under its additional model assumptions and distinct resource accounting. The evidence therefore concerns the utility of learned structure, together with the costs of finding and exploiting it.

Approximate relations and the choice of generator. The exact subspace error floor explains why hard tying can fail even with unlimited optimization within the correct base structure. Releasing weights restores expressivity and empirically recovers many such targets within the same update budget. Its deterioration at larger residuals shows that this refinement still has a finite optimization range. Source errors impose a different limitation: moderate corruption fragments the thresholded dependency extractor, while the VAE retains greater useful proposal coverage. This is an advantage over the tested extractor. Comparison with other noise-aware priors is needed to establish a broader generator-specific benefit. The two-layer tanh control supplies the complementary limit: when dense Adam is already reliable, structural search can introduce avoidable failures even with accurately recovered source assignments.

Theorem-to-experiment boundary. The dense exponential bound uses fixed coordinate variance and Gaussian sine prediction. It does not certify tuned dense baselines with fixed total norm, prove failure of arbitrary optimizers, or guarantee that VAE training learns useful coverage. The stochastic extension requires amplitude projection and a sufficiently large minibatch; the batch-64, unprojected experiments are empirical evidence for the mechanism, not direct instantiations of every theorem assumption. Likewise, the source-profile bound assumes accurate assignments, and the approximate-structure propositions do not guarantee convergence after release. The mask family's two-frequency teacher is misspecified for its student; its observed low-risk solution is not proved globally optimal. The sine codebook and two-layer tanh families are realizable.

Information, cost, and statistical scope. Equal online training-example budgets do not match FLOPs, wall-clock time, or the cost of acquiring source solutions. The original mask bank costs 4,096 target-search budgets in training-example presentations; the independent mask banks cost 4,864, 6,144, and 5,632. Codebook banks cost 0.5, 2, and 8 target-search budgets at two, four, and six blocks, respectively; each two-layer bank costs four. Prior training and validation are additional. These costs prevent a claim of overall amortized efficiency without a distribution and number of future tasks. Methods with supplied roles or blocks receive structural knowledge that source-only methods must recover.

The task families are synthetic, and only three source-bank replications support the corresponding comparisons. Many search trials share data and target functions, so their count does not measure the number of independent tasks. Reported cluster intervals are descriptive. Zero observed successes do not imply zero mathematical probability, and complete observed success does not imply reliability on every task. Broader claims would require evidence on larger combinatorial libraries, additional non-periodic optimization barriers, and complete source-to-target computational costs.

11 Conclusion

Structural proposal-and-refit search transfers information about which parameterizations are useful, while allowing their numerical values to be learned anew. Our analysis separates coverage, conditional trainability, and selection, and supplies an explicit sufficient finite-horizon separation

from a specified dense gradient method. Controlled experiments recover useful sharing relations from source solutions, distinguish structural transfer from numerical warm starts, and identify the combinatorial and identification failures that limit the gain.

Approximate sharing extends this picture: a proposed subspace can guide optimization even when its exact error floor requires subsequent unrestricted refinement. Source corruption and the two-layer tanh control delimit that benefit. Taken together, the results support structural transfer under stated optimization and coverage conditions, with VAE sampling as one implementation. They leave general generator superiority, scalable structure discovery, and end-to-end efficiency as separate questions.

Reproducibility. The accompanying `paper_vae_solver` directory contains frozen protocols, run-time source snapshots, source banks, VAE checkpoints, raw candidate arrays, aggregation scripts, and complete proofs. Experiments were completed on 25–26 October 2026. Mask tables use the corrected `matched_v2` and `replication_v2` runs. Codebook, hidden-group, robustness, and two-layer results each retain their own protocol and are not pooled as independent replications. The post-run robustness coverage diagnostic is labeled as exploratory and does not affect candidate selection. Detailed settings and execution provenance appear in Appendix G.

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A Population Risk and Dense Gradient Descent

This appendix proves the finite-horizon statements for the realizable single-frequency model in Equation (4.1). All gradient steps in this appendix use the exact population risk.

Lemma A.1 (Gaussian sine risk)**.** *For $v = e_1$, $w = (t, u)$, and*

$$b(t) = \tfrac{1}{2}\left(e^{-(t-\omega)^2/2} - e^{-(t+\omega)^2/2}\right), \tag{A.1}$$

the risk has the form in Equations (4.2) and (4.3), with $B(w) = e^{-\|u\|^2/2} b(t)$ and $|b(t)|, |b'(t)| \le 1$. Its derivatives are

$$\partial_\gamma \mathcal{L} = 2A\gamma - 2B, \tag{A.2}$$

$$\nabla_w \mathcal{L} = 2\gamma^2 e^{-2\|w\|^2} w - 2\gamma \nabla_w B, \tag{A.3}$$

$$\partial_t B = e^{-\|u\|^2/2} b'(t), \qquad \nabla_u B = -uB. \tag{A.4}$$

Proof. For a standard Gaussian input, $\mathbb{E}e^{ia^\top x} = e^{-\|a\|^2/2}$. Applying the product-to-sum identity gives

$$\mathbb{E}[\sin(a^\top x)\sin(b^\top x)] = \tfrac{1}{2}\left(e^{-\|a-b\|^2/2} - e^{-\|a+b\|^2/2}\right). \tag{A.5}$$

Expanding the squared prediction error yields the stated risk. Each exponential in b lies in $[0, 1]$. The derivative terms satisfy $|z|e^{-z^2/2} \le e^{-1/2}$, so their half-sum is less than one. Differentiating the closed form gives the remaining expressions. □

Proof of Theorem 4.1. Inside the radius-one ball about w_0, $\|u\| \ge s$ and $\|w\| \le M$. Consequently,

$$A \ge \tfrac{1}{2}(1 - e^{-2s^2}), \quad |B| \le e^{-s^2/2}, \quad \|\nabla B\| \le (1 + M)e^{-s^2/2}. \tag{A.6}$$

If $|\gamma| \le 2$, the weight-gradient norm is bounded by $G(s, M)$. The amplitude update can be written

$$\gamma^+ = (1 - 2\eta A)\gamma + 2\eta A \frac{B}{A}. \tag{A.7}$$

Because $0 < A \le 1/2$ and $\eta \le 1$, this is a convex combination. Moreover, for $s \ge 1$,

$$\left|\frac{B}{A}\right| \le \frac{2e^{-s^2/2}}{1 - e^{-2s^2}} < 2. \tag{A.8}$$

Thus the amplitude interval is invariant as long as the weights remain in the ball. Inductively, $\|w_t - w_0\| \le \eta t G \le 1$ for $t \le T$, which closes both bounds. Minimizing the quadratic risk over any amplitude then gives

$$\mathcal{L}(\gamma, w) \ge C_\omega - \frac{B^2}{A} \ge C_\omega - \frac{2e^{-s^2}}{1 - e^{-2s^2}} \ge C_\omega - 3e^{-s^2}. \tag{A.9}$$

□

Corollary A.2 (Gaussian initialization)**.** *For fixed $\sigma > 0$ and $w_0 \sim \mathcal{N}(0, \sigma^2 I_P)$, with probability at least $1 - \delta_P$,*

$$\|u_0\| \ge \sigma\sqrt{(P-1)/2}, \qquad \|w_0\| \le 2\sigma\sqrt{P}. \tag{A.10}$$

Under the conditions following Equation (4.6), the probability of risk at most $C_\omega/2$ by time T is at most δ_P per dense run. The probability that any of N runs succeeds is at most $N\delta_P$.

Proof. For $X \sim \chi^2_n$, $\mathbb{E}e^{\lambda X} = (1-2\lambda)^{-n/2}$. Exponential Markov bounds with $\lambda = -1/2$ and $1/4$ yield

$$\mathbb{P}(X < n/2) \le e^{n(1/4-\log(2)/2)} \le e^{-n/16}, \tag{A.11}$$

$$\mathbb{P}(X > 4n) \le e^{n(-1+\log(2)/2)} \le e^{-n/2}. \tag{A.12}$$

Apply the lower-tail inequality to $\|u_0\|^2/\sigma^2$ and the upper-tail inequality to $\|w_0\|^2/\sigma^2$. A union bound gives the stated event. On this event $s \ge s_P$, $M \le M_P$, and $G(s,M) \le G(s_P, M_P)$. Theorem 4.1 precludes the required risk. A second union bound covers N runs without requiring their independence. □

Lemma A.3 (A local basin on the correct coordinate)**.** *For $w = te_1$, the Hessian at $(\gamma, t) = (1, \omega)$ is*

$$H_* = \begin{pmatrix} 1 - e^{-2\omega^2} & 2\omega e^{-2\omega^2} \\ 2\omega e^{-2\omega^2} & 1 + (1-4\omega^2)e^{-2\omega^2} \end{pmatrix} \succ 0. \tag{A.13}$$

There exist $r, \mu, \Lambda > 0$, depending only on ω, such that $\mu I \preceq \nabla^2 \mathcal{L} \preceq \Lambda I$ in a radius-r ball about this solution. For $0 < \eta \le 1/\Lambda$, GD started in the ball remains there and obeys

$$\mathcal{L}(\gamma_T, t_T) \le \tfrac{1}{2}\Lambda r^2 (1-\eta\mu)^{2T}. \tag{A.14}$$

Proof. At zero residual, the Hessian is twice the Gram matrix of $\sin(\omega X)$ and $X\cos(\omega X)$. These functions are linearly independent when $\omega > 0$: the coefficients of X and X^3 in a vanishing linear combination force both combination coefficients to be zero. An almost-everywhere identity under a positive Gaussian density extends to an everywhere identity by continuity. The Gram matrix is therefore positive definite. Gaussian differentiation identities give Equation (A.13).

Continuity of the Hessian provides a sufficiently small ball and uniform spectral bounds. Relative to its center, the gradient is the integral of the Hessian along the connecting line segment. Its GD update matrix has operator norm at most $1 - \eta\mu$, so the ball is invariant and distances contract. The Λ-smoothness bound and zero risk at the center give Equation (A.14). □

The two-frequency perturbation. Adding a sufficiently small low-frequency teacher component preserves a strict local minimum near the high-frequency solution: the risk gradient depends smoothly on its amplitude, and the Hessian in Equation (A.13) is invertible. The implicit-function theorem and Hessian continuity give the claim. This argument supplies an unspecified positive perturbation range; it does not certify the actual amplitudes 0.35–0.45, establish their global optimum, or transfer the full trapping theorem without further work.

B Codebook Geometry and the Unified Subspace Theorem

Proof of Proposition 3.1. The gradient and Hessian follow from the chain rule for the linear map $c \mapsto Uc$. Multiplying each update by U gives the induced ambient update. Nonempty one-hot assignment columns have disjoint support, so $D = U^\top U$ is positive diagonal. The matrix $UD^{-1}U^\top$ is symmetric, idempotent, and has image range(U); it is the orthogonal projector. □

Lemma B.1 (Effective dimension under normalized sharing)**.** *Let $V = UD^{-1/2}$ and suppose $v \in \mathrm{range}(V)$ with $\|v\| = 1$. Writing $x' = V^\top x$ and $v' = V^\top v$ gives*

$$x' \sim \mathcal{N}(0, I_k), \qquad \|v'\| = 1, \qquad y = \sin(\omega (v')^\top x'), \qquad f = \gamma \sin(a^\top x'). \tag{B.1}$$

Thus the population objective and minibatch gradients in (γ, a) are those of the corresponding k-dimensional sine problem.

Proof. The Gaussian image has covariance $V^\top V = I_k$. Since $v = Vv'$ and $w = Va$, both teacher and student linear forms depend only on x'. The identities hold for every example, hence also for their gradients and expectations. □

Proof of Theorem 4.2. Condition on a good structure. By Lemma B.1, the problem is k-dimensional. An orthogonal change of the intrinsic coordinates sends the target to ωe_1, preserving isotropic Gaussian initialization and GD. At the exact solution, the joint amplitude/first-coordinate Hessian is H_* from Equation (A.13). Each remaining coordinate has second derivative

$$2\mathbb{E}[X_j^2 \cos^2(\omega X_1)] = 1 + e^{-2\omega^2} > 0, \qquad j \geq 2. \tag{B.2}$$

All cross terms with these coordinates vanish by independence and Gaussian symmetry. The full Hessian is positive definite.

For each $1 \leq k \leq k_0$, Hessian continuity gives a local ball with positive lower and finite upper spectral bounds. Take the minimum radius and lower bound, and the maximum upper bound, over this finite set of dimensions. They are independent of P and orientation. The contraction argument of Lemma A.3 proves local convergence under Equation (4.7). The amplitude starts at one, so the probability of starting inside the ball is the Gaussian mass in Equation (4.8). It is positive; explicitly,

$$\mathbb{P}(\|Z - \omega e_1\| \leq r) \geq \frac{r^k}{2^{k/2}\sigma^k \Gamma(k/2+1)} \exp\left(-\frac{(\omega + r)^2}{2\sigma^2}\right) > 0. \tag{B.3}$$

This is the ball volume times the minimum Gaussian density within it. Averaging over the good-structure event gives success probability at least $q\beta_*$. Independence of complete structural attempts gives $1 - (1 - q\beta_*)^N$.

For the dense comparison, rotate the full input space to send v to e_1. The initialization, Gaussian distribution, and GD are invariant under this rotation, so Corollary A.2 gives the upper bound $N\delta_P$. For fixed T and σ, the dense trapping conditions hold at sufficiently large P, while the structured local constants remain unchanged. This establishes a compatible finite-horizon separation whenever the lower bound exceeds the upper bound. □

Masks and codebooks are not universally nested. A mask with s independently trainable nonzero coordinates can be encoded using a fixed zero codeword and s private codewords. However, at a fixed small codebook size, sharing imposes equalities that a sparse mask need not impose. Conversely, a single shared nonzero value represents a dense constant vector that a small mask cannot represent. The appropriate comparison is therefore between particular representable subspaces, not an unconditional claim that either parameterization is more expressive.

Proposition B.2 (A dense target unavailable to a small mask)**.** *Let $v = P^{-1/2}(s_1, \ldots, s_P)$ with $s_i \in \{-1, 1\}$, and let $y = \sin(\omega v^\top x)$. A two-value codebook with the correct assignments has a zero-risk solution $w_i = \omega s_i/\sqrt{P}$, $\gamma = 1$. Among all predictors depending only on any fixed subset of s input coordinates, the minimum squared risk is*

$$\mathcal{L}_s^* = \tfrac{1}{2}(1 - e^{-2\omega^2}) - \tfrac{1}{2}e^{-\omega^2(1-s/P)}(1 - e^{-2\omega^2 s/P}). \tag{B.4}$$

It is strictly positive for $s < P$ and $\omega > 0$.

Proof. For an observed subset S of size s, decompose $v^\top x$ into independent observed and missing Gaussian terms. The conditional mean is

$$\mathbb{E}[y \mid x_S] = e^{-\omega^2(1-s/P)/2} \sin(\omega v_S^\top x_S). \tag{B.5}$$

This minimizes squared risk among all functions of x_S. Its second moment is the second term in Equation (B.4), while $\mathbb{E}y^2$ is the first. Conditional variance decomposition yields the result. Since the conditional mean itself is a scaled sine neuron on S, the bound is attainable in the masked student class. Missing Gaussian variance is positive when $s < P$, so the conditional variance is nonzero. The codebook representation follows by assigning each sign to the corresponding value $\pm\omega/\sqrt{P}$. □

Interpretation of normalization. The choice $c_j \sim \mathcal{N}(0, \sigma^2/n_j)$ yields $\mathbb{E}\,\|Uc\|^2 = k\sigma^2$. For unnormalized independent centroids, the same expectation is $\sigma^2 \sum_j n_j = P\sigma^2$. Reducing the number of trainable scalars alone does not control orthogonal energy. Likewise, ordinary centroid GD and group-mean GD have different induced metrics despite sharing a function class. These distinctions motivate the separate experimental ablations.

C A Sufficient Stochastic-Gradient Regime

This appendix establishes a sufficient regime for minibatch SGD with the *output amplitude projected onto* $[-2, 2]$ *after each update*. Weights are not projected. The experiments use unprojected amplitudes and batch 64, so the following results do not certify those precise implementations.

Lemma C.1 (Conditional gradient variance)**.** *For one example in Equation (4.1) and* $|\gamma| \le 2$*, let*

$$g_w = 2(\gamma \sin(w^\top x) - y)\gamma \cos(w^\top x)x, \qquad g_\gamma = 2(\gamma \sin(w^\top x) - y)\sin(w^\top x). \tag{C.1}$$

In dimension d*,*

$$\|g_w\|^2 \le 144\,\|x\|^2\,, \qquad |g_\gamma|^2 \le 36. \tag{C.2}$$

For an average of b *fresh independent examples, conditionally on the past* $\mathcal{F}_t$*, the gradient error is unbiased and satisfies*

$$\mathbb{E}[\|\widehat{g}_w - \nabla_w \mathcal{L}\|^2 \mid \mathcal{F}_t] \le \frac{144d}{b}, \qquad \mathbb{E}[\|\widehat{g} - \nabla \mathcal{L}\|^2 \mid \mathcal{F}_t] \le \frac{144d + 36}{b}. \tag{C.3}$$

Proof. The residual has absolute value at most three. Bounding the sine and cosine by one gives the pointwise inequalities. Since $\mathbb{E}\,\|x\|^2 = d$, these also bound second moments. Fresh-example independence divides the conditional variance of the average by b, and variance is bounded by the second moment. □

Theorem C.2 (Dense SGD escape probability)**.** *Let* $r_0 = \|u_0\|$*. Choose a radius* $a > 0$ *such that* $s = r_0 - a \ge 1$*, set* $M = \|w_0\| + a$*, and let* $G = G(s, M)$ *from Equation (4.4). If* $\eta TG < a$*, amplitude-projected SGD with batch* b *exits the radius-*a *weight ball by time* T *with probability at most*

$$\delta_{\text{noise}} = \min\left\{1, \frac{144d\eta^2 T}{b(a - \eta TG)^2}\right\}. \tag{C.4}$$

On the event of no exit, every risk up to T *is at least* $C_\omega - 3e^{-s^2}$*.*

Proof. Within the ball, the gradient bound in the deterministic trapping proof applies because projection keeps $|\gamma| \le 2$. Let τ be the first exit time and define the stopped cumulative noise

$$Z_t = \eta \sum_{j=0}^{t-1} \mathbf{1}\{j < \tau\}(\widehat{g}_{w,j} - \nabla_w \mathcal{L}_j). \tag{C.5}$$

This is a vector martingale. Orthogonality of its differences and Lemma C.1 imply $\mathbb{E}\|Z_T\|^2 \le 144d\eta^2 T/b$. If exit occurs by T, the accumulated deterministic drift before exit has norm at most ηTG, so $\max_{t\le T}\|Z_t\| \ge a - \eta TG$. The maximal inequality for the nonnegative submartingale $\|Z_t\|^2$ gives Equation (C.4). The risk lower bound is the same quadratic minimization used in Theorem 4.1. □

Lemma C.3 (Finite-horizon inexact-gradient coupling)**.** *Suppose exact and approximate gradient trajectories start at the same point, the gradient is L-Lipschitz throughout a convex region containing the compared points, and every gradient error has norm at most ν. For step size $\eta > 0$,*

$$\left\|\widehat{\theta}_T - \theta_T\right\| \le \frac{\nu}{L}\left[(1+\eta L)^T - 1\right]. \tag{C.6}$$

For $L = 0$, the right-hand side is $\eta T\nu$. The bound also holds with nonexpansive projection when the next exact iterate is fixed by that projection.

Proof. Subtracting updates gives $d_{t+1} \le (1+\eta L)d_t + \eta\nu$ with $d_0 = 0$. Summing the geometric recursion yields the result. Nonexpansiveness preserves the same one-step inequality under the stated projection condition. □

Proposition C.4 (Success within a good subspace)**.** *Use a local ball from the proof of Theorem 4.2, reducing its radius to $r < 1/2$ if necessary. Assume initial amplitude one and initial intrinsic weights within distance $r/2$ of the optimum. For $\eta \le 1/\Lambda$, set*

$$a_0 = \min\{r/4, \sqrt{\varepsilon/(2\Lambda)}\}, \qquad \nu = \frac{a_0\Lambda}{(1+\eta\Lambda)^T - 1}, \tag{C.7}$$

where $T \ge 1$ is large enough that exact GD reaches distance at most a_0. In intrinsic dimension k, amplitude-projected SGD attains risk at most ε with conditional probability at least

$$1 - \min\left\{1, \frac{(144k+36)T}{b\nu^2}\right\}. \tag{C.8}$$

Proof. Conditional Markov bounds applied to Equation (C.3), followed by a union bound over T steps, give failure probability at most $(144k+36)T/(b\nu^2)$ for the event that some gradient error exceeds ν. The argument may be stopped at the first prospective exit from the local ball. On the complementary event, Lemma C.3 keeps the SGD trajectory within distance a_0 of exact GD. The exact trajectory stays within radius $r/2$, so SGD stays within $3r/4$; this closes the stopped-process induction. The exact amplitude remains strictly inside $[-2, 2]$, and projection is nonexpansive. At time T the distance to the optimum is at most $2a_0$. Smoothness gives risk at most $2\Lambda a_0^2 \le \varepsilon$. □

Corollary C.5 (Strict stochastic separation)**.** *In the setting of Theorem 4.2, define $\beta_{*,1/2}$ by replacing r with $r/2$ in Equation (4.8), and use common local constants over $k \le k_0$. Then*

$$p^{\mathrm{SGD}}_{\mathrm{struct}} \ge q\beta_{*,1/2}\left[1 - \frac{(144k_0+36)T}{b\nu^2}\right]_+. \tag{C.9}$$

On the dense Gaussian concentration event, choose uniform bounds a_P, s'_P, M'_P with $G_P = G(s'_P, M'_P)$, $\eta TG_P < a_P$, and $C_\omega - 3e^{-(s'_P)^2} > \varepsilon$. Then

$$p^{\mathrm{SGD}}_{\mathrm{dense}} \le \delta_P + \frac{144P\eta^2 T}{b(a_P - \eta TG_P)^2}. \tag{C.10}$$

When the first lower bound exceeds the second upper bound, structural SGD has strictly higher success probability. For fixed $q > 0$, k_0, ω, σ, and ε, these conditions hold for some finite batch and sufficiently large P.

Proof. The lower bound is the probability of proposing a good subspace and starting in its smaller local ball, multiplied by Equation (C.8). The upper bound combines Theorem C.2 with the exceptional Gaussian probability. To see compatibility, fix the local horizon T, which does not depend on P. Choose $a_P = \sigma\sqrt{(P-1)/2}/2$ on the concentration event. Then $s'_P \geq a_P$ and one may take $M'_P = 2\sigma\sqrt{P} + a_P$. Thus G_P decays exponentially, while P/a_P^2 remains bounded. First take a sufficiently large finite b to preserve a positive structured lower bound and make the limiting dense noise correction smaller; then take P sufficiently large to control δ_P, risk, and drift. The success ordering is preserved under the same number of independent complete restarts by the monotonicity of $p \mapsto 1-(1-p)^N$. □

For masks with a single active coordinate, set $k_0 = 1$, so the variance constant is 180. These bounds are deliberately conservative, especially the exponential-in-T coupling used to establish local stochastic success. They demonstrate existence of a compatible regime, not a practical batch-size recommendation or a prediction for the empirical success rates.

D Proposal Coverage, Refitting, and Selection

Proposition D.1 (Independent complete attempts)**.** *Let $b_T(s)$ be the probability of attaining risk at most ε after training within structure s, including initialization and training noise. Let $p_Q = \mathbb{E}_{s\sim Q} b_T(s)$. For N independent complete attempts,*

$$\mathbb{P}(\textit{an accurate candidate exists}) = 1-(1-p_Q)^N. \tag{D.1}$$

If $Q(\mathcal{G}) \geq q$ and $b_T(s) \geq \beta$ for $s \in \mathcal{G}$, this probability is at least $1-(1-q\beta)^N$.

Proof. Each attempt fails with probability $1-p_Q$; independence multiplies these probabilities. Restricting the expectation defining p_Q to $\mathcal{G}$ gives $p_Q \geq q\beta$. □

Shared minibatch streams generally invalidate unconditional independence. If independence holds only conditionally on a common stream, the formula must first be evaluated conditionally and then averaged. Substituting the unconditional mean into the power is generally incorrect.

Proposition D.2 (Two-stage refitting)**.** *Suppose exploration selects a structure in $\mathcal{G}$ with probability at least ρ_K. Conditional on the full exploration history and such a structure, assume that R fresh refits are independent and each succeeds with probability at least β_T. Then the probability that at least one accurate candidate exists is at least*

$$\rho_K[1-(1-\beta_T)^R]. \tag{D.2}$$

If the final selection procedure discards all accurate candidates when one exists with unconditional probability at most δ, the returned-success probability is at least $[\rho_K(1-(1-\beta_T)^R)-\delta]_+$.

Proof. Condition on each exploration history selecting a good structure. The probability of all refits failing is at most $(1-\beta_T)^R$. Integrating over those histories proves the first statement. Subtract the final selection failure event to obtain the second. □

For K IID proposals, the probability of *encountering* $\mathcal{G}$ is $1-(1-Q(\mathcal{G}))^K$. It need not equal ρ_K because unsuccessful weight optimization can obscure a good structure during exploration.

Proposition D.3 (Independent final selection)**.** *Suppose H predictors are constructed independently of a final sample of size n, and their pointwise losses lie in $[0,B]$. With probability at least $1-\delta$, all empirical risks satisfy*

$$|\widehat{\mathcal{L}}_h - \mathcal{L}_h| \leq t_n, \qquad t_n = B\sqrt{\frac{\log(2H/\delta)}{2n}}. \tag{D.3}$$

If one predictor has risk at most $\varepsilon - 2t_n$, empirical minimization returns a predictor with risk at most ε on this event.

Proof. Hoeffding's inequality applies to each fixed predictor; a union bound covers all H. The risk of the empirical minimizer is at most its empirical risk plus t_n, which is at most the best true risk plus $2t_n$. □

This proposition is not directly applicable to the experimental reuse of selection data, because second-stage predictors depend on that data through the chosen structure. Moreover, experimental amplitudes are unbounded. The reported sine results instead use exact population risk for evaluation.

Proposition D.4 (VAE coverage from reconstruction and aggregation)**.** *Let P be a source distribution over structures, $q_\phi(z \mid s)$ an encoder, $Q_Z = \int q_\phi(z \mid s)\, dP(s)$ its aggregated posterior, and P_Z the sampling prior. For the same deterministic discretized decoder D used in search, suppose*

$$\mathbb{P}_{s\sim P,\, z\sim q_\phi(\cdot|s)}(D(z) \neq s) \leq e, \qquad \mathrm{KL}(Q_Z \| P_Z) \leq \kappa. \tag{D.4}$$

Writing $Q = D_{\#}P_Z$, for any good set $\mathcal{G}$,

$$Q(\mathcal{G}) \geq P(\mathcal{G}) - e - \sqrt{\kappa/2}. \tag{D.5}$$

If $\mathrm{TV}(P_{\mathrm{new}}, P) \leq \Delta$, one also has

$$Q(\mathcal{G}) \geq P_{\mathrm{new}}(\mathcal{G}) - \Delta - e - \sqrt{\kappa/2}. \tag{D.6}$$

Proof. Jointly sampling a source structure and its reconstruction couples P with $D_{\#}Q_Z$, so their total variation distance is at most e. Deterministic pushforward contracts total variation. Pinsker's inequality gives $\mathrm{TV}(Q_Z, P_Z) \leq \sqrt{\kappa/2}$. The triangle inequality proves the first bound, and an additional Δ proves the second. □

Convexity provides the potentially looser bound

$$\mathrm{KL}(Q_Z \| P_Z) \leq \mathbb{E}_{s\sim P}\, \mathrm{KL}(q_\phi(z \mid s) \| P_Z). \tag{D.7}$$

Nevertheless, weighted reconstruction losses, straight-through penalties, and a beta-VAE objective do not automatically certify the stated discrete reconstruction event or target coverage. For disjoint source and target supports of whole structures, $\Delta = 1$ and the transfer bound can be vacuous.

Proposition D.5 (Composition from independent parts)**.** *If latent sampling and decoding factorize into J independent parts, and part j generates its required component with probability at least a_j, then the required complete structure has probability at least $\prod_{j=1}^{J} a_j$. Under the corresponding partwise assumptions of Proposition D.4, one may take*

$$a_j = \left[P_j(\{s_j^*\}) - e_j - \sqrt{\kappa_j/2} \right]_+ . \tag{D.8}$$

Proof. Independence multiplies component probabilities; apply the preceding coverage bound to each component. □

The shared decoder and global top-two discretization in the mask VAE do not enforce this exact factorization. Its partwise objective encourages the behavior, so the proposition is a sufficient condition rather than an established checkpoint property. The codebook block-sampling procedure does use independent latent samples across blocks.

Proposition D.6 (Dependence can change coverage exponentially)**.** *Suppose J categorical components each take h values and the desired distribution is uniform on the h diagonal structures, where all components are equal. A generator reproducing the joint distribution assigns mass h^{-1} to each diagonal structure. The product of the exact marginals assigns mass h^{-J}, a ratio of h^{J-1}.*

Proof. The joint distribution has h equally weighted atoms. Each marginal is uniform on h values, and their product multiplies J factors of $1/h$. □

A jointly modeling VAE can represent this dependence, but the proposition does not guarantee successful VAE training. Replay of all diagonal structures also attains $1/h$, as the empirical control confirms.

E When Source Solutions Identify Hidden Components

This elementary analysis explains one sufficient condition for learning proposal support in the follow-up experiment. It assumes that the source assignments have already been recovered correctly. It is not a guarantee for learning arbitrary decompositions or for VAE training, and we make no novelty claim for the underlying profile-collision argument.

Model. An assignment vector has constant coordinates and J nonempty groups of varying coordinates. A coordinate in group j equals $Z_j \oplus s_i$, where $s_i \in \{0, 1\}$ is fixed and $\oplus$ denotes exclusive-or. The groups and signs are unknown and may be arbitrarily permuted. Across source tasks, each Z_j is a fair bit, and every distinct pair (Z_j, Z_l) is independent. Full joint independence is unnecessary. Observe m independent source assignment vectors. Group empirically constant coordinates together and connect two remaining coordinates when their profiles agree or disagree on at least a fraction $1 - \rho$ of the observations, for $0 \leq \rho < 1/2$. Return the connected components.

Proposition E.1 (Recovery from signed source profiles)**.** *With $r = \lfloor \rho m \rfloor$, the algorithm recovers exactly the J varying groups, with constant coordinates in a separate fixed component, except on an event of probability at most*

$$\delta_m = \min\left\{1,\ J2^{1-m} + J(J-1)2^{-m} \sum_{l=0}^{r} \binom{m}{l}\right\}. \tag{E.1}$$

For exact matching, $\rho = 0$, this becomes $\delta_m \leq \min\{1, J(J+1)2^{-m}\}$.

Proof. A given varying factor is constant across all m observations with probability 2^{1-m}. A union bound over the J factors gives the first term. On the complementary event, empirically constant coordinates are exactly the truly constant coordinates. Within each varying group, profiles are identical or complementary, so every pair of its coordinates is connected.

For two different factors, their disagreement indicators across independent tasks are independent fair bits. Their disagreement count is therefore Binomial$(m, 1/2)$. The probability of at most r disagreements or at most r agreements is $2^{1-m} \sum_{l=0}^{r} \binom{m}{l}$; the two events are disjoint since $r < m/2$. Fixed coordinate signs can only exchange these two events. A union bound over the $\binom{J}{2}$ pairs gives the second term. If none of these events occurs, there is no edge between different true groups, so connected components recover the varying partition exactly. □

Finite source banks. The experiment subsamples accepted source rows without replacement. If the full source bank has M rows and each factor and each pairwise disagreement are balanced, the same proof replaces each binomial tail by its hypergeometric counterpart. Writing

$$H_r(M,m) = \sum_{l=0}^{r} \frac{\binom{M/2}{l}\binom{M/2}{m-l}}{\binom{M}{m}}, \tag{E.2}$$

with out-of-range binomial coefficients interpreted as zero, the failure bound is

$$\min\{1, 2JH_0(M,m) + J(J-1)H_r(M,m)\}. \tag{E.3}$$

For a complete even-parity bank with $J \geq 3$, the required pairwise balance holds, including when every combination is repeated the same number of times. The complete bank is recovered exactly for $\rho < 1/2$. Incorrectly recovered source assignments violate this premise; their probability must be added to the bound rather than ignored.

Corollary E.2 (Coverage after successful recovery)**.** *On successful recovery, uniform independent choice between the two observed patterns of each varying component assigns mass 2^{-J} to every factor combination, including combinations absent from the source support. Suppose each correct assignment, with fresh continuous initialization, yields an accurate candidate with probability at least β. For N independent proposals and conditional fits, the probability that the candidate set contains an accurate solution is at least*

$$(1-\delta_m)\left[1-(1-2^{-J}\beta)^N\right]. \tag{E.4}$$

Proof. Successful recovery implies that both patterns of each varying factor were observed, and that patterns of distinct factors can be combined independently. The uniform product distribution therefore covers all 2^J combinations equally. Conditional on the recovered partition, each independent proposal and fit succeeds with probability at least $2^{-J}\beta$. Multiply the conditional success bound by the recovery probability. □

Remark E.3 (What the coverage statement does not imply). The corollary concerns existence of an accurate candidate. Returning it also requires a reliable selection procedure. It does not certify the empirical selection and shared minibatches used in the experiments. Empirical component frequencies, rather than uniform choices, give target mass $\prod_j \widehat{p}_j(z_j^\star)$; this is the distribution used by the component sampler in the study. Even with perfect identification, uniform coverage decreases exponentially in J. Learning the decomposition alone does not eliminate combinatorial search.

A source-identifiability failure. For $J = 2$, the even-parity support is only $\{(0,0),(1,1)\}$, so $Z_1 = Z_2$ on every source task. The pairwise-independence premise fails. All varying profiles are identical up to complement and merge into one component. That component can reproduce only the observed joint patterns; independent recombination cannot produce the held-out odd combinations. Distinct hidden partitions can explain the same source observations while allowing different unseen combinations, so source data alone do not identify the intended partition without additional assumptions. This is a limitation of source-only identification, not an impossibility result for algorithms that also search using target observations.

F Approximate Sharing: Irreducible Error and Stability

The following elementary Gaussian calculation separates representation error from optimization failure when the target only approximately obeys a proposed sharing structure. It does not require a learned generator.

Proposition F.1 (Exact error floor for a fixed subspace)**.** *Let $x \sim \mathcal{N}(0, I_D)$, let S be a linear subspace with orthogonal projector P, and write $w_\star = u + r$, where $u = Pw_\star$ and $r = (I - P)w_\star$. Set $A = \|u\|^2$ and $B = \|r\|^2$. Among all square-integrable predictors measurable with respect to Px, the minimum squared risk for the target $\sin(w_\star^\top x)$ is*

$$R_S^* = \frac{1}{2}\left[1 - e^{-2(A+B)} - e^{-B}(1 - e^{-2A})\right]. \tag{F.1}$$

The minimum is attained by the tied sine predictor $e^{-B/2}\sin(u^\top x)$, so it is also the exact optimum over $\{\gamma \sin(w^\top x) : w \in S, \gamma \in \mathbb{R}\}$.

Proof. Orthogonal Gaussian projections are independent. Conditional on Px, $r^\top x$ is centered Gaussian with variance B. Therefore

$$\mathbb{E}[\sin(w_\star^\top x) \mid Px] = \sin(u^\top x)\mathbb{E}[\cos(r^\top x)] = e^{-B/2}\sin(u^\top x).$$

Conditional expectation minimizes squared risk over measurable predictors. Its risk is the second moment of the target minus the second moment of the conditional expectation. Since $\mathbb{E}\sin^2(v^\top x) = (1 - e^{-2\|v\|^2})/2$, this difference is exactly Equation (F.1). The displayed minimizer belongs to the tied sine class, proving equality for that smaller class as well. □

For the correct base assignment in the robustness experiment, S is spanned by its two group indicators, A is approximately 16, and $B = \epsilon^2$. Thus R_S^* is extremely close to $(1 - e^{-\epsilon^2})/2$. At residual norms 0.3, 0.6, 1, and 2, this gives approximately 0.043, 0.151, 0.316, and 0.491, respectively. Even an ideal hard-tied optimizer cannot reach the 0.1 threshold in the last three cases for this base assignment. An arbitrary alternative assignment has its own projector and error floor; the statement does not presume that every wrong assignment has larger risk. Releasing all weights removes this particular representation restriction, but does not guarantee that a finite optimization trajectory finds the unrestricted optimum.

Proposition F.2 (Risk stability under target perturbations)**.** *For any fixed predictor $f \in L^2(\mathcal{N}(0, I_D))$, let $R_v(f) = \mathbb{E}[(f(x) - \sin(v^\top x))^2]$. Then, for any $v, r \in \mathbb{R}^D$,*

$$\sqrt{R_{v+r}(f)} \le \sqrt{R_v(f)} + \|r\|. \tag{F.2}$$

Consequently, a predictor with $R_v(f) \le \eta$ retains risk at most $(\sqrt{\eta} + \epsilon)^2$ whenever $\|r\| \le \epsilon$.

Proof. Apply the triangle inequality in L^2 and the one-Lipschitz property of sine:

$$\left\|\sin((v + r)^\top x) - \sin(v^\top x)\right\|_{L^2} \le \left\|r^\top x\right\|_{L^2} = \|r\|.$$

□

This is a statement about fixed predictors and persistence of low-risk regions. It neither bounds the error of the learned assignment extractor nor transfers a convergence guarantee to a perturbed SGD trajectory.

G Extended Experimental Results and Reproduction Protocol

G.1 Mask splits, training, and tuning

Let $(a, b) \in \{0, 1, 2, 3\}^2$ index the two relevant within-role coordinates. Source pairs satisfy $(b - a) \bmod 4 \in \{0, 1\}$, validation pairs have residue two, and evaluation pairs have residue three. The original source functions use seeds 0/1. Dense tuning uses function seed 400; primary

Table 8: Descriptive paired differences in the primary mask experiment. Intervals are 95% percentile bootstrap intervals from 10,000 resamples of eight function/family clusters, conditional on the source bank. Units are percentage points.

Partwise VAE minus control	Difference	Interval
Empirical product	2.1	$[-2.9, 7.3]$
Uniform known roles	-2.6	$[-8.1, 3.6]$
Enumeration	-13.5	$[-19.8, -6.5]$
Dense Adam	40.6	$[27.1, 52.6]$
Dense SGD	65.9	$[60.2, 70.8]$

Table 9: Mask scaling success rates (%); 192 searches per method and dimension. Extra coordinates are known to be irrelevant for the structural priors. Fan-in SGD uses standard deviation $2\sqrt{8/d}$ and the selected SGD rate. Oracle mask success is approximately 100% across dimensions.

d	VAE	Product	Uniform edges	SGD	Momentum	Adam	Fan-in SGD
8	67.2	68.8	14.1	1.6	5.2	27.1	1.0
16	61.5	66.1	3.1	0.0	0.0	13.5	0.0
32	60.9	71.4	1.0	0.0	0.0	0.0	0.0
64	65.6	71.4	0.0	0.0	0.0	0.0	0.0

evaluation uses 401/402. VAE seeds are 4213, 4214, and 4215. Original checkpoint hashes are recorded before evaluation.

The mask VAE has a 16-dimensional latent vector, a Neural Functional encoder, and a shared decoder with two width-128 GELU hidden layers. The partwise objective uses two eight-dimensional latent groups: reconstruction of each output column replaces the other group with independent Gaussian noise. Hard decoding selects the top two logits globally. The encoder has column-equivariant layers with local and mean-context terms. This design encourages partwise behavior without enforcing independent hard-decoded parts. Full source snapshots and training recipes accompany the checkpoints.

The empirical product prior balances source tasks when estimating component frequencies. Whole-mask replay samples training-role elite masks uniformly. Validation-role cards do not enter these priors. All generators are frozen throughout target searches.

For dense tuning, SGD rates are $\{0.05, 0.2\}$, momentum rates $\{0.01, 0.05\}$ with coefficient 0.9, and Adam rates $\{0.01, 0.05\}$ with $(\beta_1, \beta_2) = (0.9, 0.999)$ and numerical epsilon 10^{-8}. Coordinate standard deviations are $\{0.5, 1.25, 2.0\}$. Each configuration is evaluated with four 64-start searches on each of four validation tasks. At $d = 8$, the selected (rate, standard deviation) pairs are $(0.2, 1.25)$, $(0.05, 1.25)$, and $(0.05, 1.25)$ for SGD, momentum, and Adam, respectively. Sparse training uses rate 0.05 and active-coordinate standard deviation two.

The two-output MSE gradients are evaluated analytically on each minibatch; they are the same gradients as automatic differentiation of the empirical prediction loss. Initial output amplitudes equal one and are trainable. Selection uses 4,096 separate examples. The first 32 sparse fits cycle four initialization vectors, followed by 32 new initializations on the selected mask. Enumeration visits 16 masks in random order and shifts its initialization chain on the second pass so that each mask receives two different starts. Dense and oracle mask baselines receive 64 distinct initializations.

Table 10: Allocation at 64 fits per mask search; 192 searches per row and method. The four exploratory initialization chains are reused at every allocation, including $64 + 0$.

Exploration + refits	VAE success	Product success
4+60	16.7%	19.3%
16+48	44.8%	51.0%
32+32	67.2%	68.8%
64+0	24.0%	24.0%

Table 11: Independent mask banks. Each rate uses 128 searches. Source costs count training-example presentations relative to one 64-fit online search; VAE training is additional.

	Bank 751	Bank 752	Bank 753
Enumeration success (%)	82.8	84.4	80.5
Role-uniform success (%)	66.4	71.9	68.8
Product success (%)	71.9	71.1	57.0
VAE success (%)	65.6	66.4	60.9
Dense Adam success (%)	32.0	40.6	35.9
Whole-mask replay success (%)	0.0	0.0	0.0
Source fits	311,296	393,216	360,448
Accepted elites	349	390	386
Equivalent online budgets	4,864	6,144	5,632

G.2 Additional mask results

Independent banks use seeds 751/752/753, new within-role coordinate permutations, source functions 501/502, and evaluation functions 601/602. Source masks are uniform among all two-edge masks. Fits are collected in blocks of 16,384, with at most eight blocks per source task and a stopping target of at least ten training and two validation elites. The 80/20 training/validation role is assigned before filtering by measured MSE below 0.1. Each bank trains one new partwise VAE with the original recipe. Table 11 shows per-bank evaluation rates and source costs.

G.3 Population, dependence, and tanh controls

The population experiment uses one sine output, frequency four, 8,192 initializations per setting, 4,096 exact GD steps, rate 0.2, and float64. Dense fixed-scale initialization has standard deviation two; fan-in uses $2\sqrt{8/d}$. The correct mask retains a single relevant coordinate with standard deviation two. These are per-initialization results, distinct from the multi-fit solver success rates.

The dependence control supplies 2,048 training and 512 validation diagonal structures directly, with eight categories and $J = 2, 4, 8$ components. It trains three categorical VAEs (seeds 871/872/873), using latent dimension eight, width-128 GELU encoder/decoder hidden layers, 1,500 Adam steps, batch 256, rate 10^{-3}, and KL weight 0.05. The checkpoint is selected by validation objective. Coverage is measured on 32,768 new prior codes. The source contains all eight diagonal structures, and replay is an exact strong control.

The tanh experiment uses two realizable outputs $y_j = \tanh(h_j x_{r_j})$, function seed 701, and dimensions eight and 32. Frozen sine-trained VAEs and the same 32+32 budget are retained. Dense SGD uses rate 0.2 and standard deviation 1.25; sparse methods use 0.05 and two. There is no tanh-specific retuning. Each method has 96 searches per dimension, and final risk uses 65,536 independent examples.

Table 12: Population-GD control. Success rates and certified-trapping fractions are percentages. The final column checks the sufficient radius-one trapping condition of Theorem 4.1, not a separate certificate for MSE < 0.1.

d	Dense fixed scale	Dense fan-in	Correct mask	Certified trapping
2	43.63	32.92	59.23	0.61
8	1.33	1.33	58.76	38.00
16	0.00	0.05	58.83	94.27
32	0.00	0.00	58.36	100.00
64	0.00	0.00	58.62	100.00
128	0.00	0.00	58.85	100.00

Table 13: Representation-only dependence control: average probability mass on each required diagonal structure. This is not a target optimization or unseen-structure-transfer experiment.

Parts	Joint VAE	Empirical product	Replay
2	0.12326	0.0157	0.12500
4	0.12171	0.00025	0.12500
8	0.12010	6.68×10^{-8}	0.12500

G.4 Codebook protocol and source costs

The frozen codebook protocol uses bank seeds 9101/9102/9103 and source functions 810/811. Source fits use four dense starts, 2,048 SGD steps, batch 64, rate 0.1, and initial coordinate standard deviation $1/\sqrt{P}$. Acceptance is based only on empirical MSE below 0.005. The sign of the output amplitude is absorbed into the weights before clustering; two centroids are fitted by 20 Lloyd iterations.

The block VAE encodes the one-hot representation of eight binary assignments through a width-128 GELU hidden layer to a four-dimensional Gaussian latent. Its decoder has a width-128 GELU hidden layer and 16 output logits, decoded by per-position argmax. Training uses 1,500 Adam steps, batch 128, rate 10^{-3}, and KL weight 0.05. The final checkpoint is used without target selection. Independent latent samples generate the target blocks.

Validation uses function 900 and up to four even-parity assignment patterns at high target frequencies. Evaluation uses functions 901/902 and all two odd-parity patterns at $J = 2$, all eight at $J = 4$, and eight predetermined patterns out of 32 at $J = 6$. Each bank/function/pattern context has 16 trials. Dense SGD rates are $\{0.05, 0.2\}$; Adam rates are $\{0.005, 0.02, 0.05\}$; the root-mean-square norm parameters are $\{1, \sqrt{8}, 4\}$, implemented as coordinate standard deviation equal to that parameter divided by $\sqrt{P}$. Codebook learning uses rate 0.05 and normalized-coordinate standard deviation two. Initial amplitudes are one and are not projected.

Both label assignments I and their global complements are treated as equivalent in cor-

Table 14: Negative control on the realizable tanh family. Rates use 96 searches per cell; exact counts are included.

Method	$d = 8$	$d = 32$
Dense SGD	100.0% (96/96)	96.9% (93/96)
Uniform known roles	91.7% (88/96)	91.7% (88/96)
Empirical product	85.4% (82/96)	85.4% (82/96)
Partwise VAE	83.3% (80/96)	83.3% (80/96)

Table 15: Codebook source-bank construction, identical counts for each of the three seeds at a given block count. Exact recovered assignments are a diagnostic, not an acceptance criterion. Source cost excludes VAE training.

Blocks	Accepted tasks	Source fits	Target-budget equivalents	Exact recovery
2	4/4	16	0.5	100%
4	16/16	64	2.0	100%
6	64/64	256	8.0	100%

Table 16: Codebook success rates (%) by independent source bank. Per-bank denominators are 64 searches for two blocks and 256 for four or six blocks.

Method	Blocks	Seed 9101	Seed 9102	Seed 9103
VAE assignments	2	90.6	90.6	79.7
VAE assignments	4	49.2	52.0	46.1
VAE assignments	6	12.5	13.7	12.1
Empirical blocks	2	90.6	87.5	82.8
Empirical blocks	4	52.0	46.5	41.4
Empirical blocks	6	14.1	18.4	19.9
Dense Adam	2	28.1	31.2	40.6
Dense Adam	4	0.8	1.2	1.6
Dense Adam	6	0.0	0.0	0.0

rectness diagnostics. The raw-initialization ablation changes only centroid scaling, and the summed-gradient ablation changes only the update metric. Failed or nonfinite fits are counted as failures; none occurred in this codebook series.

G.5 Artifacts and provenance

Each run saves its plan, source snapshots, seeds, and raw NPZ arrays of candidate losses and selected results. The codebook suite additionally stores learned source weights, recovered assignments, centroid values, generator checkpoints, and assignment-coverage diagnostics. Temporary codebook runs were exported to `runs/codebook_v1`; their final use does not depend on retaining the temporary storage.

Primary mask tables use `matched_v2` and `replication_v2`. The first enumeration implementation repeated the same initialization on its second pass; the corrected version shifts the initialization chain. The matched comparison was recomputed, and replication evaluation was recomputed using the same saved banks and generators. Earlier versions remain for audit and are not extra replications. Scaling and allocation did not include enumeration and were unaffected.

The implementation uses PyTorch 2.6.0+cu118 on V100 GPUs and one CPU thread per experiment process. Methods are evaluated in joint vectorized batches, so recorded context times are not isolated per-method timings. Final raw-data auditing reproduced the primary and independent-bank mask counts. All 24,192 codebook searches agree with empirical-loss candidate selection, and all 42 method/dimension aggregates agree with the published summaries. This audit did not perform additional model training.

G.6 Robustness and two-layer follow-ups

The protocols are `robustness_protocol.json` and `twolayer_protocol.json`; the accompanying `ROBUSTNESS_AND_TWOLAYER_PROTOCOL.md` specifies perturbations, budgets, source acquisition, and

selection. The new target frequency seeds are 921/922 for robustness; the two-layer numerical-coefficient seeds are 941/942, with 930 reserved for validation. No original evaluation seed is reused as a new independent replication.

For the single-sine study, dense SGD uses learning rate 0.2 and Adam 0.02, both with expected initial squared weight norm eight, as selected by the original independent validation. Tied and released SGD use learning rate 0.05 and normalized codebook coordinates of standard deviation two. Actual early source collection uses four dense restarts per each of 16 tasks, with 8, 32, or 128 updates; these acquisition costs are 1/128, 1/32, and 1/8 of one target search budget, respectively. The 2,048-update reference uses the original accepted bank and costs two such budgets. Noise conditions reuse that bank and do not reduce its acquisition cost.

For the two-layer study, validation selects initial hidden-weight norm 0.5 for both dense optimizers, with learning rates 0.2 for SGD and 0.02 for Adam. Tied SGD selects norm 4 and rate 0.2. Tuning evaluates 27 settings across two validation tasks with eight starts per setting. All structural methods share the tied setting; numerical transfer uses the dense Adam rate. Every VAE in both studies has a full-vector binary categorical decoder, latent dimension eight, 128-unit hidden layers, 3,000 Adam updates, learning rate 0.001, batch size 128, and KL weight 0.01. It is frozen before proposal sampling, and target losses do not update the prior.

Raw arrays are exported to `runs/robustness_v1` and `runs/twolayer_v1`. The report script `robustness_twolayer_report.py` audits all 10,848 selected indices against empirical argmin selection and verifies returned errors and success counts. It writes a per-search CSV, JSON summaries, source diagnostics, plots, and Table 7. Confidence intervals in the JSON and auxiliary tanh plot resample whole task/source-bank contexts with 5,000 bootstrap draws. Pooled replicate counts are not treated as independent draws from a task population. The post-run coverage diagnostic evaluates only stored proposals and is reproducible from the same report script.

The first robustness attempt encountered a result-filename error and was excluded. Corrected workers use the same numerical protocol. Work was split by source bank; the first worker was stopped after completing its assigned bank, and any duplicate work on a subsequent bank was excluded from export. Each exported shard records its selected bank and original source snapshot. Neither the failed attempt nor duplicate work is counted as a replication.